

WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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Abstract.

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1. Introduction

Definition 1.1. Let $1 \leq r < \infty$. For a measurable set $E \subset \mathbb{R}^N$ of finite positive measure and $f \in L^r(E)$, define

$$\text{osc}_r(f; E) := \left(\frac{1}{|E|} \int_E |f - (f)_E|^r dx \right)^{1/r}.$$

Definition 1.2. Let $0 < p < \infty$, $0 < q \leq \infty$, and $1 \leq r < \infty$. Define

$$OSC^{p,q}(\mathbb{R}^N) := \left\{ f \in L^r_{\text{loc}}(\mathbb{R}^N) : \|f\|_{OSC^{p,q}(\mathbb{R}^N)} < \infty \right\},$$

where

$$\|f\|_{OSC^{p,q}(\mathbb{R}^N)} := \sum_{v=1}^{2^N} \left\| \{\text{osc}_r(f; Q)\}_{Q \in \mathbb{D}^v} \right\|_{\ell^{p,q}}.$$

2010 Mathematics Subject Classification. 47B10, 42B20, 43A85.

Key words and phrases. Schatten class, Riesz transform commutators, Dunkl operator, Besov space, Nearly weakly orthogonal sequence.

By [11, Proposition 4.1], the above definition is independent of r .

1.1. Notation. Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N , p , and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \leq CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda \ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \leq p \leq \infty$, we denote by p' the conjugate of p , which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a measurable set E by

$$(f)_{E,dx} := \frac{1}{|E|} \int_E f(x) dx.$$

For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

2. Preliminaries

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane $v^\perp$ orthogonal to v is defined by

$$\sigma_v x := x - 2 \frac{\langle x, v \rangle}{\|v\|^2} v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $\|v\|^2 = 2$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R . Let $k : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper. We will denote by $\mathcal{O}(x) = \{\sigma(x) : \sigma \in G\}$ the G -orbit of a point $x \in \mathbb{R}^N$, and $\mathcal{O}(E) = \bigcup_{x \in E} \mathcal{O}(x)$ the G -orbit of $E \subset \mathbb{R}^N$.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{k(v)} dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v \in R} k(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^{\mathbf{N}} \prod_{v \in R} (|\langle x, v \rangle| + r)^{k(v)},$$

where the implicit constant only depends on N, R and k . It follows from (1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^{\mathbf{N}} \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

[Add the definition of A_∞ . Zhijie will do it.]

Lemma 2.1. Let

$$dw(x) = \varpi(x) dx, \quad \varpi(x) = \prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)}.$$

Then $\varpi \in A_\infty(\mathbb{R}^N, dx)$.

Proof. The assertion is immediate if $k \equiv 0$. Otherwise, choose $p > 1$ sufficiently large so that

$$\frac{2|R_+|k(v)}{p-1} < 1, \quad v \in R_+.$$

For every $0 < \beta < 1$, $v \in R_+$, and Euclidean ball $B(x_0, r)$, we apply an orthogonal change of variables to deduce that

$$\begin{aligned} \int_{B(x_0, r)} |\langle x, v \rangle|^{-\beta} dx &\lesssim \frac{1}{r|v|} \int_{\langle x_0, v \rangle - |v|r}^{\langle x_0, v \rangle + |v|r} |s|^{-\beta} ds \\ &= (|v|r)^{-\beta} \int_{\frac{\langle x_0, v \rangle}{|v|r} - 1}^{\frac{\langle x_0, v \rangle}{|v|r} + 1} |s|^{-\beta} ds \\ &\lesssim (|v|r)^{-\beta} \left(1 + \frac{|\langle x_0, v \rangle|}{|v|r} \right)^{-\beta} \\ &\lesssim (|\langle x_0, v \rangle| + r)^{-\beta}. \end{aligned}$$

Hence, by Hölder's inequality,

$$\begin{aligned} \int_{B(x_0, r)} \varpi^{-\frac{1}{p-1}} dx &= \int_{B(x_0, r)} \prod_{v \in R_+} |\langle x, v \rangle|^{-\frac{2k(v)}{p-1}} dx \\ &\leq \prod_{v \in R_+} \left(\int_{B(x_0, r)} |\langle x, v \rangle|^{-\frac{2|R_+|k(v)}{p-1}} dx \right)^{\frac{1}{|R_+|}} \\ &\lesssim \prod_{v \in R_+} (|\langle x_0, v \rangle| + r)^{-\frac{2k(v)}{p-1}}. \end{aligned}$$

On the other hand, by (1) and the G -invariance of k ,

$$\int_{B(x_0, r)} \varpi dx = \frac{w(B(x_0, r))}{|B(x_0, r)|} \lesssim \prod_{v \in R_+} (|\langle x_0, v \rangle| + r)^{2k(v)}.$$

Therefore,

$$\left(\int_B \varpi dx \right) \left(\int_B \varpi^{-\frac{1}{p-1}} dx \right)^{p-1} \lesssim 1$$

uniformly over all Euclidean balls B . Thus $\varpi \in A_p(\mathbb{R}^N, dx)$, and hence $\varpi \in A_\infty(\mathbb{R}^N, dx)$. $\square$

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{k(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v \in R} k(v) \delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x \rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$, $j = 1, 2, \dots, N$.

Definition 2.2. The Weyl chambers are defined as the closures of connected components of the set

$$(3) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}.$$

Define $d(x, y) := \min_{\sigma \in G} \|x - \sigma(y)\|$ as the distance between two G -orbits $\mathcal{O}(x)$ and $\mathcal{O}(y)$. Recall from [6, Chapter VII, proof of Theorem 2.12] that

$$(4) \quad d(x, y) = \|x - y\| \text{ if and only if } x, y \in \Gamma$$

for some closed Weyl chamber Γ .

It is clear that $\mathcal{O}(B(x, r)) = \{y \in \mathbb{R}^N : d(x, y) < r\}$. Moreover,

$$w(B(x, r)) \leq w(\mathcal{O}(B(x, r))) \leq |G|w(B(x, r))$$

for all $x \in \mathbb{R}^N$ and $r > 0$. We also denote $B^d(x, r) := \{y \in \mathbb{R}^N : d(x, y) < r\}$, i.e., $B^d(x, r) = \mathcal{O}(B(x, r))$.

Denote by $K_j(x, y)$ the integral kernel of $R_{\text{Dunkl},j}$. For every $x, y \in \mathbb{R}^N$ and $r > 0$, let

$$V(x, y, r) := \max\{w(B(x, r)), w(B(y, r))\}.$$

The following lemma establishes derivative estimates for the Dunkl-Riesz transform kernel.

Lemma 2.3. Assume that $N \geq 1$. Then for every pair of multi-indices α, β , every $j \in \{1, \dots, N\}$, and every $x, y \in \mathbb{R}^N$ with $d(x, y) > 0$,

$$(5) \quad \left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl},j}(x, y) \right| \lesssim_{\alpha, \beta, R, k} \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-(|\alpha|+|\beta|)}}{V(x, y, d(x, y))}.$$

Proof. Case 1. Suppose that $|\alpha| + |\beta| = 0$. By [4, Theorem 1.1],

$$|K_{\text{Dunkl},j}(x, y)| \lesssim \frac{d(x, y)}{\|x - y\|} \frac{1}{w(B(x, d(x, y)))}.$$

By the definition of d , the G -invariance of dw , and (2),

$$V(x, y, d(x, y)) \lesssim w(B(x, d(x, y))),$$

and hence (5) follows.

Case 2. Suppose that $|\alpha| + |\beta| \geq 1$. By [?, Theorem 3.1],

$$\left| \partial_x^\alpha \partial_y^\beta h_t(x, y) \right| \lesssim_{\alpha, \beta, R, k} t^{-\frac{|\alpha|+|\beta|}{2}} \left(1 + \frac{\|x - y\|}{\sqrt{t}} \right)^{-2} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t}.$$

Recall from the proof of Lemma 2.4 that

$$K_{\text{Dunkl},j}(x, y) = -\frac{C}{2} \int_0^\infty (y_j - x_j) h_t(x, y) \frac{dt}{t^{3/2}}.$$

Since

$$\partial_x^\alpha \partial_y^\beta ((y_j - x_j) h_t(x, y)) = (y_j - x_j) \partial_x^\alpha \partial_y^\beta h_t(x, y) - \alpha_j \partial_x^{\alpha-e_j} \partial_y^\beta h_t(x, y) + \beta_j \partial_x^\alpha \partial_y^{\beta-e_j} h_t(x, y),$$

where the corresponding term is omitted if $\alpha_j = 0$ or $\beta_j = 0$, we obtain

$$\begin{aligned} \left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl},j}(x, y) \right| &\lesssim \frac{1}{\|x - y\|} \int_0^\infty t^{-\frac{|\alpha|+|\beta|+1}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x,y)^2/t} dt \\ &\quad + \frac{1}{\|x - y\|^2} \int_0^\infty t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x,y)^2/t} dt. \end{aligned}$$

By (2),

$$V(x, y, \sqrt{t})^{-1} \lesssim V(x, y, d(x, y))^{-1} \begin{cases} \left(\frac{d(x, y)}{\sqrt{t}} \right)^N, & 0 < t \leq d(x, y)^2, \\ \left(\frac{d(x, y)}{\sqrt{t}} \right)^N, & t \geq d(x, y)^2. \end{cases}$$

Hence,

$$\begin{aligned} &\int_0^\infty t^{-\frac{|\alpha|+|\beta|+1}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x,y)^2/t} dt \\ &\lesssim \frac{1}{V(x, y, d(x, y))} \left(d(x, y)^N \int_0^{d(x,y)^2} t^{-\frac{N+|\alpha|+|\beta|+1}{2}} e^{-cd(x,y)^2/t} dt + d(x, y)^N \int_{d(x,y)^2}^\infty t^{-\frac{N+|\alpha|+|\beta|+1}{2}} dt \right) \\ &\lesssim \frac{d(x, y)^{1-|\alpha|-|\beta|}}{V(x, y, d(x, y))}. \end{aligned}$$

The first integral in the parentheses is estimated by the change of variables $s = d(x, y)^2/t$, while the second one converges since $N + |\alpha| + |\beta| > 1$.

Case 2.1. Suppose that $N + |\alpha| + |\beta| > 2$. We apply the same argument to see that

$$\begin{aligned} &\int_0^\infty t^{-\frac{|\alpha|+|\beta|}{2}} V(x, y, \sqrt{t})^{-1} e^{-cd(x,y)^2/t} dt \\ &\lesssim \frac{1}{V(x, y, d(x, y))} \left(d(x, y)^N \int_0^{d(x,y)^2} t^{-\frac{N+|\alpha|+|\beta|}{2}} e^{-cd(x,y)^2/t} dt + d(x, y)^N \int_{d(x,y)^2}^\infty t^{-\frac{N+|\alpha|+|\beta|}{2}} dt \right) \\ &\lesssim \frac{d(x, y)^{2-|\alpha|-|\beta|}}{V(x, y, d(x, y))}. \end{aligned}$$

Since $d(x, y) \leq \|x - y\|$, it follows that

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl},j}(x, y) \right| \lesssim_{\alpha, \beta, R, k} \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-(|\alpha|+|\beta|)}}{V(x, y, d(x, y))}.$$

Case 2.2. Suppose that $N = 1$ and $|\alpha| + |\beta| = 1$. For the terms in which a derivative falls on $y_j - x_j$, retaining the factor $(1 + \|x - y\|/\sqrt{t})^{-2}$ and splitting the integral at

$d(x, y)^2$ and $\|x - y\|^2$, we obtain from (2)

$$\begin{aligned} & \int_0^\infty t^{-3/2} \left(1 + \frac{\|x - y\|}{\sqrt{t}}\right)^{-2} V(x, y, \sqrt{t})^{-1} e^{-cd(x, y)^2/t} dt \\ & \lesssim \frac{d(x, y)^N}{\|x - y\|^2 V(x, y, d(x, y))} \int_0^{d(x, y)^2} t^{-\frac{N+1}{2}} e^{-cd(x, y)^2/t} dt \\ & \quad + \frac{d(x, y)}{\|x - y\|^2 V(x, y, d(x, y))} \int_{d(x, y)^2}^{\|x - y\|^2} \frac{dt}{t} + \frac{d(x, y)}{V(x, y, d(x, y))} \int_{\|x - y\|^2}^\infty \frac{dt}{t^2} \\ & \lesssim \frac{1}{\|x - y\| V(x, y, d(x, y))}, \end{aligned}$$

where we used

$$\frac{d(x, y)}{\|x - y\|} \log \frac{\|x - y\|}{d(x, y)} \lesssim 1.$$

Combining this estimate with the preceding estimate for the first term, we obtain

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}, j}(x, y) \right| \lesssim_{\alpha, \beta, R, k} \frac{1}{\|x - y\| V(x, y, d(x, y))} = \frac{d(x, y)}{\|x - y\|} \frac{d(x, y)^{-1}}{V(x, y, d(x, y))}.$$

This completes the proof of Lemma 2.3. $\square$

Lemma 2.4. For $j \in \{1, 2, \dots, N\}$, and for every ball $B = B(x_0, r) \subset \mathbb{R}^N$, there is another ball $\widehat{B} = B(y_0, r)$ with $y_0 = x_0 + 5re_j$ such that for every $(x, y) \in B \times \widehat{B}$, $K_j(x, y)$ does not change sign and satisfies

$$|K_j(x, y)| \geq \frac{C}{w(B(x_0, r))}.$$

Proof. The lower bound is given in [4, Theorem 1.2], while the fact that $K_j(x, y)$ does not change sign can be seen in the proof of [4, Theorem 1.2]. Indeed, with the choice $y_0 := x_0 + 5re_j$, we have $y_j - x_j \geq 3r$ for all $(x, y) \in B \times \widehat{B}$. Together with the fact that the heat kernel $h_t(x, y)$ associated with the Dunkl Laplacian is positive, the representation

$$K_j(x, y) = -\frac{C}{2} \int_0^\infty \frac{y_j - x_j}{t} h_t(x, y) \frac{dt}{\sqrt{t}},$$

implies that $K_j(x, y) < 0$ (and thus, does not change sign) on $B \times \widehat{B}$. $\square$

2.2. Systems of dyadic cubes.

Definition 2.5. Let $X \subseteq \mathbb{R}^N$ be a nonempty open set. A countable family $\mathcal{Q} := \bigcup_{k \in \mathbb{Z}} \mathcal{Q}_k$, $\mathcal{Q}_k := \{Q_\alpha^k : \alpha \in \mathcal{A}_k\}$, of Borel sets $Q_\alpha^k \subseteq X$ is called a system of dyadic cubes over X if it has the following properties:

(i) For each $k \in \mathbb{Z}$, we have

$$X = \bigcup_{\alpha \in \mathcal{A}_k} Q_\alpha^k;$$

(ii) If $\ell \geq k$, then either $Q_\beta^\ell \subseteq Q_\alpha^k$ or $Q_\alpha^k \cap Q_\beta^\ell = \emptyset$;

(iii) For each (k, α) and each $\ell \leq k$, there exists a unique β such that $Q_\alpha^k \subseteq Q_\beta^\ell$;

(iv) For each (k, α) there exist 2^N indices β such that

$$Q_\beta^{k+1} \subseteq Q_\alpha^k, \text{ and } Q_\alpha^k = \bigcup_{\substack{Q \in \mathcal{D}_{k+1} \\ Q \subseteq Q_\alpha^k}} Q;$$

(v) For each (k, α) , one has

$$B_X(x_\alpha^k, A_1 2^{-k}) \subseteq Q_\alpha^k \subseteq B_X(x_\alpha^k, A_2 2^{-k}) =: B(Q_\alpha^k)$$

for some positive constants A_1, A_2 depending only on the geometry of X , where we set $B_X(x, r) := B(x, r) \cap X$ for all $x \in X$ and $r > 0$.

(vi) If $\ell \geq k$ and $Q_\beta^\ell \subseteq Q_\alpha^k$, then

$$B(Q_\beta^\ell) \subseteq B(Q_\alpha^k).$$

Each Q_α^k is called a dyadic cube of generation k , with center $x_\alpha^k \in Q_\alpha^k$ and side-length $\ell(Q_\alpha^k) = 2^{-k}$.

Notation 2.6. For every dyadic cube $Q \in \mathcal{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

Notation 2.7. Let

$$\mathbb{D}^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^0$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube is a product of half-open intervals

$$[m_1 2^{-k}, (m_1 + 1) 2^{-k}) \times \cdots \times [m_N 2^{-k}, (m_N + 1) 2^{-k}),$$

with $(m_1, \dots, m_N) \in \mathbb{Z}^N$. Moreover, we let

$$\mathbb{D}_\Gamma := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma, k}$$

be the system of dyadic cubes over a fixed chamber Γ , whose construction will be given in Corollary 3.7.

Lemma 2.8. There exist 2^N systems of dyadic cubes $\mathbb{D}^1, \mathbb{D}^2, \dots, \mathbb{D}^{2^N}$ such that for any ball $B = B(x_B, r_B) \subset \mathbb{R}^N$ with $\frac{2^{-k-2}}{3} \leq r_B < \frac{2^{-k-1}}{3}$ for some $k \in \mathbb{Z}$, there exists a cube

$$Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}_k^\nu \text{ such that}$$

$$B \subset Q \subset 12 \sqrt{N} B.$$

Proof. • Step 1. Recall the one-dimensional setting given in [9, Section 2].

Let $\mathbb{D}^{(0)}(\mathbb{R})$ be the standard system of dyadic intervals over $\mathbb{R}$ (see Notation 2.7). Let $\delta = 1/3$ and define

$$\mathbb{D}^{(\delta)}(\mathbb{R}) = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\delta)}(\mathbb{R}),$$

where for $k \geq 0$,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \{I + \delta \mid I \in \mathbb{D}_k^{(0)}(\mathbb{R})\},$$

while for $k < 0$ and k even,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \left\{ \left\lfloor \frac{m}{2^k} \right\rfloor + \delta + \sum_{j=(k/2)+1}^0 2^{-2j}, \frac{m+1}{2^k} + \delta + \sum_{j=(k/2)+1}^0 2^{-2j} \right\} \mid m \in \mathbb{Z}.$$

These choices together with the nestedness property completely determine the collections $\mathbb{D}_k^{(\delta)}(\mathbb{R})$ for $k < 0$, k odd.

Fix an interval $J \subset \mathbb{R}$ with $2^{-k-1}/3 \leq |J| < 2^{-k}/3$ for some $k \in \mathbb{Z}$. Now we set $A_k = \{m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for every fixed k and

- (1) $A_k^{(\delta)} = \{\delta + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k \geq 0$,
- (2) $A_k^{(\delta)} = \{\delta + \sum_{j=(k/2)+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k even, and
- (3) $A_k^{(\delta)} = \{\delta + \sum_{j=(k-1)/2+1}^0 2^{-2j} + m \cdot 2^{-k} \mid m \in \mathbb{Z}\}$ for $k < 0$, k odd.

Note that for any two different points $a, b \in A_k \cup A_k^{(\delta)}$, we have $|a - b| \geq 2^{-k}/3 > |J|$. Thus, there is at most one element of $A_k \cup A_k^{(\delta)}$ contained in J . Hence, $J \cap A_k = \emptyset$ or $J \cap A_k^{(\delta)} = \emptyset$. Therefore, J must be contained in some dyadic interval $I \in \mathbb{D}_k^{(0)}$ or $I \in \mathbb{D}_k^{(\delta)}$ and $|I| = 2^{-k} \leq 6|J|$.

• Step 2. We extend the construction to $\mathbb{R}^N$ ($N \geq 1$).

For $\vartheta = (\vartheta_1, \vartheta_2, \dots, \vartheta_N) \in \{0, \delta\}^N$, define

$$\mathbb{D}^{(\vartheta)} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k^{(\vartheta)}, \quad \mathbb{D}_k^{(\vartheta)} := \left\{ \prod_{j=1}^N I_j : I_j \in \mathbb{D}_k^{(\vartheta_j)} \right\}.$$

Applying Step 1 to $\pi_i(B)$, where $\pi_i(B)$ denotes the i -th coordinate projection, we obtain $I_i \in \mathbb{D}_k^{(0)}$ or $I_i \in \mathbb{D}_k^{(\delta)}$ with $\pi_i(B) \subset I_i$ and $|I_i| = 2^{-k} \leq 12r_B$. Let $\vartheta_i \in \{0, \delta\}$ be the corresponding shift so that $I_i \in \mathbb{D}_k^{(\vartheta_i)}$, set $\vartheta := (\vartheta_1, \dots, \vartheta_N)$, and let $Q = \prod_{j=1}^N I_j \in \mathbb{D}_k^{(\vartheta)}$. It follows that $B \subset Q$ and $\ell(Q) = 2^{-k} \leq 12r_B$. Hence, for every $y \in Q$,

$$\|y - x_B\| \leq \sqrt{N}\ell(Q) \leq 12\sqrt{N}r_B.$$

This implies that $Q \subset B(x_B, 12\sqrt{N}r_B)$. Enumerating $\{\mathbb{D}^{(\vartheta)} : \vartheta \in \{0, \delta\}^N\}$ as $\mathbb{D}^1, \dots, \mathbb{D}^{2^N}$, we complete the proof of Lemma 2.8. $\square$

2.3. Nearly weakly orthonormal sequences. Rochberg and Semmes [11] introduced the notion of nearly weakly orthonormal (NWO) sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [3, 7, 10, 1, 2, 8, 11, 13, 12]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [7, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N, \|\cdot\|, w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Let $\mathcal{Q}$ be a system of dyadic cubes. A sequence of functions $\{e_Q\}_{Q \in \mathcal{Q}}$ is an NWO sequence if

- (1) $\text{supp}(e_Q) \subseteq cQ$ for some positive constant c independent of Q
- (2) $\|e_Q\|_{L^s(\mathbb{R}^N, dw)} \leq w(Q)^{1/s-1/2}$ for some $s > 2$.

[We need to replace the following Lemma by Hytonen's version on space of homogeneous type. Zhijie Will do it later]

Lemma 2.9. Let $1 < p < \infty$, $1 \leq q \leq \infty$, and let $\mathcal{Q}$ be a system of dyadic cubes over $\mathbb{R}^N$. If

$$T \in S^{p,q}(L^2(\mathbb{R}^N, dw)),$$

then

$$\left\| \left\{ \langle Te_Q, f_Q \rangle_{L^2(\mathbb{R}^N, dw)} \right\}_{Q \in \mathcal{D}} \right\|_{\ell^{p,q}} \leq C_{N,p,q,w} \|T\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))},$$

where $\{e_Q\}_{Q \in \mathcal{D}}$ and $\{f_Q\}_{Q \in \mathcal{D}}$ are NWO sequences.

Proof. This property can be found in [11, (1.10)] (see also [7, Section 7]). $\square$

Lemma 2.10. Let $0 < p < \infty$, $0 < q \leq \infty$, and let $\mathcal{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. Assume that

$$A = \sum_{Q \in \mathcal{D}} \lambda_Q \langle \cdot, e_Q \rangle_{L^2(\mathbb{R}^N, dw)} f_Q,$$

where $\{e_Q\}_{Q \in \mathcal{D}}$ and $\{f_Q\}_{Q \in \mathcal{D}}$ are NWO sequences and $\{\lambda_Q\}_{Q \in \mathcal{D}}$ is a sequence of scalars. We have

$$(6) \quad \|A\|_{S^{p,q}} \leq C_{N,p,q,w} \|\{\lambda_Q\}_{Q \in \mathcal{D}}\|_{\ell^{p,q}}.$$

Proof. This property can be found in [11, (1.9)] (see also [7, Section 7]). $\square$

3. Analytic tools for Weyl chamber geometry

Convention. Throughout this paper, we will use the following notation:

- Let $R \subset \mathbb{R}^N$ be a finite normalized reduced root system;
- Let $W := \text{span}(R)$ with $N_0 := \dim W$;
- Let $\Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$;
- Let Γ_+ be a fixed open Weyl chamber (see Definition 3.1) and Γ be its closure.

Definition 3.1. [5, Definition 2.7] A connected component of

$$(1) \quad \{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber. Define the corresponding sets of positive and negative roots by

$$R_+ := \{\alpha \in R : \langle x, \alpha \rangle > 0 \text{ for all } x \in \Gamma_+\}, \quad R_- := -R_+.$$

Definition 3.2. [5, Definition 2.9] A root $\alpha \in R_+$ is called simple in R_+ if it cannot be written as

$$\alpha = \lambda_1 \alpha_1 + \lambda_2 \alpha_2 \quad \text{with } \lambda_1, \lambda_2 \geq 1 \text{ and } \alpha_1, \alpha_2 \in R_+.$$

The set $\Delta \subset R_+$ of all simple roots is called the simple system.

Lemma 3.3. With the notation given in Definition 3.2, Δ is a basis of W . If we denote by $\Delta = \{\alpha_1, \alpha_2, \dots, \alpha_{N_0}\}$, then the open and closed chambers can be written as

$$(2) \quad \Gamma_+ = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle > 0, i = 1, \dots, N_0\},$$

$$(3) \quad \Gamma = \{x \in \mathbb{R}^N : \langle x, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\},$$

and every root $\alpha \in R$ admits a unique expansion

$$\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$$

such that either $c_i \geq 0$ for all i (when $\alpha \in R_+$), or $c_i \leq 0$ for all i (when $\alpha \in R_-$). In particular, the simple system Δ is uniquely determined by Γ_+ (equivalently, by Γ).

Proof. Let $\Delta = \{\alpha_1, \dots, \alpha_n\} \subset R_+$ be the set of simple roots in R_+ . By [5, Proposition 2.10], every $\alpha \in R_+$ is a nonnegative linear combination of $\alpha_1, \dots, \alpha_n$. Since $R = R_+ \cup R_-$ (this follows from the fact that for each $\alpha \in R$, the continuous function $x \mapsto \langle x, \alpha \rangle$ does not vanish on the connected set Γ_+) and $R_- = -R_+$, it follows that $R \subset \text{span}(\Delta)$ and hence,

$$W = \text{span}(R) \subset \text{span}(\Delta).$$

The reverse inclusion $\text{span}(\Delta) \subset W$ is immediate from $\Delta \subset R \subset W$, so $\text{span}(\Delta) = W$. On the other hand, it follows from [5, Theorem 2.16] that the simple roots are linearly independent. Consequently, Δ is a basis of W , and therefore $n = \dim W = N_0$. Moreover, every $\alpha \in R$ has a unique expansion $\alpha = \sum_{i=1}^{N_0} c_i \alpha_i$. If $\alpha \in R_+$, the coefficients can be chosen nonnegative by [5, Proposition 2.10]. If $\alpha \in R_- = -R_+$, then $-\alpha \in R_+$ has an expansion with all coefficients nonnegative, and multiplying by -1 gives an expansion of α with all coefficients nonpositive. The descriptions (2) and (3) follow from [5, Corollary 2.11].

To see that Δ is uniquely determined by Γ_+ , note that Γ_+ determines R_+ by Definition 3.1, and the set of simple roots in R_+ is characterized intrinsically by Definition 3.2. Finally, since Γ_+ is the (unique) interior component of Γ , specifying Γ determines Γ_+ and vice versa. This completes the proof of Lemma 3.3. $\square$

Notation 3.4. With the notation given in Lemma 3.3, denote by

$$\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$$

the simple system associated with Γ .

Lemma 3.5. Each Weyl chamber Γ admits the decomposition

$$\Gamma = K(\Gamma) \times W^\perp, \quad K(\Gamma) := \{w \in W : \langle w, \alpha \rangle \geq 0 \text{ for all } \alpha \in \Delta\}.$$

Proof. Write $x = w + z$ with $w \in W$ and $z \in W^\perp$. Since $\alpha_i \in W$, we have $\langle x, \alpha_i \rangle = \langle w, \alpha_i \rangle$ for all $i = 1, \dots, N_0$. This, together with (3), implies that $x \in \Gamma$ if and only if $w \in K(\Gamma)$. This implies that $\Gamma = K(\Gamma) \times W^\perp$, and thus completes the proof. $\square$

Corollary 3.6. Let $\Phi_{W^\perp}$ be any linear isomorphism from $W^\perp$ to $\mathbb{R}^{N-N_0}$, and let

$$\begin{aligned} \Phi_W : W &\longrightarrow \mathbb{R}^{N_0}, & \Phi_W(w) &:= (\langle w, \alpha_1 \rangle, \dots, \langle w, \alpha_{N_0} \rangle), \\ \Phi : \mathbb{R}^N = W \oplus W^\perp &\longrightarrow \mathbb{R}^{N_0} \oplus \mathbb{R}^{N-N_0} \cong \mathbb{R}^N, & \Phi(w + z) &:= (\Phi_W(w), \Phi_{W^\perp}(z)). \end{aligned}$$

Then Φ is a linear isomorphism and

$$\Phi(\Gamma) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

Proof. By Lemma 3.5 we have $\Gamma = K(\Gamma) \times W^\perp$, where

$$K(\Gamma) = \{w \in W : \langle w, \alpha_i \rangle \geq 0, i = 1, \dots, N_0\}.$$

Since Δ is a basis of W , the Gram matrix $G = (\langle \alpha_i, \alpha_j \rangle)_{i,j=1}^{N_0}$ is nonsingular. Hence, $\Phi_W : W \longrightarrow \mathbb{R}^{N_0}$ is a linear isomorphism. Moreover, for $w \in W$ we have $w \in K(\Gamma)$ if and only if $\Phi_W(w) \in \mathbb{R}_+^{N_0}$, so

$$\Phi_W(K(\Gamma)) = \mathbb{R}_+^{N_0}.$$

By construction, Φ is a linear isomorphism (as a direct sum of isomorphisms), and therefore

$$\Phi(\Gamma) = \Phi(K(\Gamma) \times W^\perp) = \Phi_W(K(\Gamma)) \times \Phi_{W^\perp}(W^\perp) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0},$$

which proves the desired conclusion. $\square$

Corollary 3.7. Let Γ be an arbitrary Weyl chamber. There exists a system of dyadic cubes over Γ .

Proof. Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism from Corollary 3.6 so that

$$\Phi(\Gamma) = \Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

For each $k \in \mathbb{Z}$, define $\mathbb{D}_{\Omega,k}^0$ to be the family of half-open cubes

$$Q = \prod_{i=1}^N I_i, \quad I_i = [m_i 2^{-k}, (m_i + 1) 2^{-k}),$$

where $m_i \in \mathbb{Z}_+$ for $1 \leq i \leq N_0$ and $m_i \in \mathbb{Z}$ for $N_0 < i \leq N$. Set

$$\mathbb{D}_{\Omega}^0 := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Omega,k}^0.$$

It is standard that $\mathbb{D}_{\Omega}^0$ is a system of dyadic cubes over Ω .

Next, for each $k \in \mathbb{Z}$, define

$$\mathbb{D}_{\Gamma,k} := \{\Phi^{-1}(Q) : Q \in \mathbb{D}_{\Omega,k}^0\}, \quad \mathbb{D}_{\Gamma} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma,k}.$$

Since $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a linear isomorphism, we see that $\mathbb{D}_{\Gamma}$ is a system of dyadic cubes over Γ . $\square$

Notation 3.8. Let Φ be the linear isomorphism given in Corollary 3.6. For every Borel set $E \subset \Omega$, define

$$(4) \quad \mu(E) := w(\Phi^{-1}(E)).$$

Lemma 3.9. We have

$$(5) \quad d\mu(u, z) = \phi(u) du dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)}.$$

where for each $v \in R_+$, the coefficients $a_{v,i} \geq 0$ are uniquely determined by the simple-root expansion $v = \sum_{i=1}^{N_0} a_{v,i} \alpha_i$ and satisfy $(a_{v,1}, \dots, a_{v,N_0}) \neq \mathbf{0}$.

Proof. Since $\Phi(\Gamma) = \Omega$, we have $\Phi^{-1}(\Omega) = \Gamma$. Hence, for every nonnegative Borel function f on Ω ,

$$\int_{\Omega} f(u, z) d\mu(u, z) = \int_{\Gamma} f(\Phi(x)) dw(x).$$

We make the linear change of variables $(u, z) = \Phi(x)$, so that $x = \Phi^{-1}(u, z)$. Since Φ is linear and invertible, the Jacobian is constant and

$$dx = |\det(\Phi)|^{-1} du dz.$$

It follows from the definition of Φ that $u_i = \langle x, \alpha_i \rangle$ for $i = 1, \dots, N_0$. This implies that

$$(6) \quad \langle x, v \rangle = \left\langle x, \sum_{i=1}^{N_0} a_{v,i} \alpha_i \right\rangle = \sum_{i=1}^{N_0} a_{v,i} \langle x, \alpha_i \rangle = \sum_{i=1}^{N_0} a_{v,i} u_i.$$

Substituting these identities into the previous integral, we obtain

$$\begin{aligned} \int_{\Omega} f(u, z) d\mu(u, z) &= \int_{\Gamma} f(\Phi(x)) \left(\prod_{v \in R_+} |\langle x, v \rangle|^{2k(v)} \right) dx \\ &= \int_{\Omega} f(u, z) |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{i=1}^{N_0} a_{v,i} u_i \right)^{2k(v)} du dz. \end{aligned}$$

Therefore, (5) holds. This completes the proof of Lemma 3.9. $\square$

Lemma 3.10. Let Γ_1 and Γ_2 be two Weyl chambers. Then there exist an integer $M \geq 0$ and $\alpha_1, \dots, \alpha_M \in R$ such that, if we set $\Gamma^{(0)} := \Gamma_2$ and

$$\Gamma^{(k)} := \sigma_{\alpha_k}(\Gamma^{(k-1)}), \quad k = 1, \dots, M,$$

then the following hold:

- (1) $\alpha_k \in \Delta(\Gamma^{(k-1)})$ for each $k = 1, \dots, M$;
- (2) $\Gamma^{(M)} = \Gamma_1$.

Proof. If $\Gamma_1 = \Gamma_2$, we take $M = 0$ and there is nothing to prove. Assume $\Gamma_1 \neq \Gamma_2$. Since the Weyl group G acts transitively on the set of Weyl chambers (see [5, Theorem 2.12 (2)]), there exists $g \in G$ such that $\Gamma_1 = g(\Gamma_2)$.

Let $\Delta(\Gamma_2) = \{\beta_1, \beta_2, \dots, \beta_{N_0}\}$ be the simple system associated with Γ_2 . By [5, Theorem 2.12(4)], the Weyl group G is generated by the simple reflections $\sigma_{\beta_1}, \sigma_{\beta_2}, \dots, \sigma_{\beta_{N_0}}$. Hence, there exist indices $i_1, \dots, i_M \in \{1, \dots, N_0\}$ such that

$$g = \sigma_{\beta_{i_1}} \sigma_{\beta_{i_2}} \cdots \sigma_{\beta_{i_M}}.$$

Define

$$w_0 := \text{Id}, \quad w_k := \sigma_{\beta_{i_1}} \cdots \sigma_{\beta_{i_k}} \quad (k = 1, \dots, M),$$

and set

$$\Gamma^{(k)} := w_k(\Gamma_2) \quad (k = 0, 1, \dots, M).$$

It follows that $\Gamma^{(0)} = \Gamma_2$ and $\Gamma^{(M)} = w_M(\Gamma_2) = g(\Gamma_2) = \Gamma_1$, which implies the statement (2).

For each $k = 1, \dots, M$, define

$$\alpha_k := w_{k-1}(\beta_{i_k}) \in R.$$

The rest of the proof is divided into two steps.

- Step 1. Verify that $\Gamma^{(k)} = \sigma_{\alpha_k}(\Gamma^{(k-1)})$.

Since each $w \in G$ is orthogonal, for any root $\beta \in R$ one has the conjugation identity

$$w \sigma_{\beta} w^{-1} = \sigma_{w(\beta)}, \quad w \in G.$$

Applying this with $w = w_{k-1}$ and $\beta = \beta_{i_k}$ yields

$$\sigma_{\alpha_k} = \sigma_{w_{k-1}(\beta_{i_k})} = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}.$$

Therefore,

$$\sigma_{\alpha_k}(\Gamma^{(k-1)}) = w_{k-1} \sigma_{\beta_{i_k}} w_{k-1}^{-1}(w_{k-1}(\Gamma_2)) = w_{k-1} \sigma_{\beta_{i_k}}(\Gamma_2) = w_k(\Gamma_2) = \Gamma^{(k)}.$$

- Step 2. Verify that $\alpha_k \in \Delta(\Gamma^{(k-1)})$.

Recall that if Γ is a Weyl chamber associated with simple system $\Delta(\Gamma)$, then

$$\Gamma = \{x \in \mathbb{R}^N : \langle x, \gamma \rangle \geq 0 \text{ for all } \gamma \in \Delta(\Gamma)\}.$$

Applying $w_{k-1} \in G$ to the above description for Γ_2 and using $\langle w_{k-1}x, \beta \rangle = \langle x, w_{k-1}^{-1}\beta \rangle$, we obtain

$$\Gamma^{(k-1)} = w_{k-1}(\Gamma_2) = \{x \in \mathbb{R}^N : \langle x, w_{k-1}(\beta_j) \rangle \geq 0 \text{ for } j = 1, \dots, N_0\}.$$

By the uniqueness of the simple system associated with a Weyl chamber (see Lemma 3.3), we conclude that

$$\Delta(\Gamma^{(k-1)}) = \Delta(w_{k-1}(\Gamma_2)) = w_{k-1}(\Delta(\Gamma_2)) = \{w_{k-1}(\beta_1), \dots, w_{k-1}(\beta_{N_0})\}.$$

In particular, $\alpha_k = w_{k-1}(\beta_{i_k}) \in \Delta(\Gamma^{(k-1)})$, proving the first statement (1).

This completes the proof of Lemma 3.10. $\square$

Definition 3.11. We say that $R' \subset \Omega$ is an axis-parallel rectangle if

$$R' := I_1 \times \dots \times I_N,$$

with $I_i = [I_i^-, I_i^+] \subset [0, \infty)$ for $1 \leq i \leq N_0$ and $I_i = [I_i^-, I_i^+] \subset \mathbb{R}$ for $i > N_0$.

Denote by

$$B_\Omega(x_0, r) := \{x \in \Omega : \|x - x_0\| < r\}.$$

The following lemma demonstrates that $(\Omega, \|\cdot\|, d\mu)$ is a doubling metric measure space.

Lemma 3.12. There exists a constant $C \geq 1$ such that for every $x_0 = (u_0, z_0) \in \Omega$ and every $r > 0$,

$$(7) \quad C^{-1} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq \mu(B_\Omega(x_0, r)) \leq C r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where

$$L_v(u) := \sum_{i=1}^{N_0} a_{v,i} u_i, \quad L_{v,0} := L_v(u_0) = \sum_{i=1}^{N_0} a_{v,i} (u_0)_i.$$

In particular, there exists a constant $C_{\text{dbl}, \Omega} \geq 1$ such that for every $x \in \Omega$ and every $r > 0$,

$$\mu(B_\Omega(x, 2r)) \leq C_{\text{dbl}, \Omega} \mu(B_\Omega(x, r)).$$

Proof. Fix $x_0 = (u_0, z_0) \in \Omega$ and $r > 0$. For each $v \in R_+$ define the linear form

$$L_v(u) := a_v \cdot u, \quad a_v := (a_{v,1}, \dots, a_{v,N_0}) \in \mathbb{R}_+^{N_0} \setminus \{0\}.$$

Since R_+ is finite and each $a_v \neq 0$, we have

$$L_* := \max_{v \in R_+} \|a_v\| < \infty, \quad m_* := \min_{v \in R_+} \sum_{i=1}^{N_0} a_{v,i} > 0.$$

Upper bound. Let $x = (u, z) \in B_\Omega(x_0, r)$. Then

$$\|u - u_0\| \leq \|x - x_0\| \leq r.$$

Hence, for every $v \in R_+$,

$$|L_v(u) - L_{v,0}| = |a_v \cdot (u - u_0)| \leq \|a_v\| \cdot \|u - u_0\| \leq L_* r.$$

Since $L_v(u) \geq 0$ on Ω (all coefficients and coordinates are nonnegative), we obtain

$$L_v(u) \leq L_{v,0} + L_* r \leq C_1 (L_{v,0} + r), \quad C_1 := \max\{1, L_*\}.$$

Therefore,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \leq C_1^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_2 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

Integrating over $B_\Omega(x_0, r)$ and using $|B_\Omega(x_0, r)| \leq |B(x_0, r)| = \omega_N r^N$, where ω_N is the volume of the unit ball in $\mathbb{R}^N$, gives

$$\mu(B_\Omega(x_0, r)) = \int_{B_\Omega(x_0, r)} \phi(u) dx \leq C_2 |B(x_0, r)| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

with $C_3 := C_2 \omega_N$.

Lower bound. We construct a smaller Euclidean ball B_c contained in $B_\Omega(x_0, r)$ on which ϕ admits a uniform positive lower bound in terms of $\prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}$. To this end, we first choose

$$\lambda := 1 + \frac{L_*}{m_*} > 1, \quad \eta := \frac{1}{4(\sqrt{N_0} \lambda + 1)}, \quad \theta := \lambda \eta.$$

These parameters depend only on L_* and m_* , and satisfy

$$\theta > \eta > 0, \quad \sqrt{N_0} \theta + \eta = \eta(\sqrt{N_0} \lambda + 1) = \frac{1}{4} < 1.$$

Let $\mathbf{1}_{N_0} := (1, \dots, 1) \in \mathbb{R}^{N_0}$ and define the center

$$x_c := (u_c, z_c) := (u_0 + \theta r \mathbf{1}_{N_0}, z_0), \quad B_c := B(x_c, \eta r).$$

We claim that $B_c \subset B_\Omega(x_0, r)$. To see this inclusion, we first apply the triangle inequality to see that for any $x \in B_c$,

$$\|x - x_0\| \leq \|x - x_c\| + \|x_c - x_0\| \leq \eta r + \theta r \sqrt{N_0} \leq r,$$

since $\sqrt{N_0} \theta + \eta < 1$. Hence $B_c \subset B(x_0, r)$. Moreover, for any $x = (u, z) \in B_c$, we have $\|u - u_c\| \leq \|x - x_c\| \leq \eta r$. This implies that for each $1 \leq i \leq N_0$,

$$u_i \geq (u_c)_i - \eta r = (u_0)_i + (\theta - \eta)r \geq 0,$$

since $u_0 \in \mathbb{R}_+^{N_0}$ and $\theta \geq \eta$. Thus $u \in \mathbb{R}_+^{N_0}$ and $x \in \Omega$. Therefore, $B_c \subset B_\Omega(x_0, r)$. This finishes the proof of the claim.

Next we obtain a uniform lower bound for $L_v(u)$ on B_c . Write any $x = (u, z) \in B_c$ as

$$u = u_0 + \theta r \mathbf{1}_{N_0} + \Delta_u, \quad z = z_0 + \Delta_z, \quad \|\Delta_u\|^2 + \|\Delta_z\|^2 \leq (\eta r)^2,$$

so in particular $\|\Delta_u\| \leq \eta r$. Then for all $x = (u, z) \in B_c$ and all $v \in R_+$,

$$\begin{aligned} L_v(u) &= a_v \cdot u = a_v \cdot u_0 + \theta r a_v \cdot \mathbf{1}_{N_0} + a_v \cdot \Delta_u \\ &\geq L_{v,0} + \theta r \sum_{i=1}^{N_0} a_{v,i} - \|a_v\| \|\Delta_u\| \geq L_{v,0} + r K_0 \geq c_0 (L_{v,0} + r). \end{aligned}$$

with

$$K_0 := \theta m_* - \eta L_* = \eta(\lambda m_* - L_*) = \eta m_* > 0, \quad c_0 := \min\{1, K_0\} \in (0, 1].$$

Consequently, for all $u \in B_c$,

$$\phi(u) = \prod_{v \in R_+} L_v(u)^{2k(v)} \geq c_0^{2 \sum_{v \in R_+} k(v)} \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} =: C_4 \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)}.$$

This, in combination with the fact that $B_c \subset B_\Omega(x_0, r)$, yields

$$\mu(B_\Omega(x_0, r)) \geq \mu(B_c) = \int_{B_c} \phi(u) dx \geq C_4 |B_c| \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = C_5 r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)},$$

where $C_5 := C_4 \omega_N \eta^N > 0$.

The second statement follows immediately from the first one. Indeed, by the first statement,

$$\mu(B_\Omega(x_0, 2r)) \leq C_3 (2r)^N \prod_{v \in R_+} (L_{v,0} + 2r)^{2k(v)} \leq C_3 2^{N+\mathcal{K}} r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} \leq C_3 C_5^{-1} 2^{N+\mathcal{K}} \mu(B_\Omega(x_0, r)),$$

where we set $\mathcal{K} := 2 \sum_{v \in R_+} k(v)$. Therefore, $(\Omega, \|\cdot\|, \mu)$ is doubling with the doubling constant $C_{\text{dbl}, \Omega} := C_3 C_5^{-1} 2^{N+\mathcal{K}}$. This finishes the proof of Lemma 3.12. $\square$

Let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be the linear isomorphism given in Corollary 3.6 such that $\Phi(\Gamma) = \Omega$. Since Φ is linear and invertible, there exist positive constants c_Φ and C_Φ such that for any $x, y \in \mathbb{R}^N$,

$$(8) \quad c_\Phi \|x - y\| \leq \|\Phi(x) - \Phi(y)\| \leq C_\Phi \|x - y\|.$$

Denote by $B_\Gamma(x, r) := \{y \in \Gamma : \|y - x\| < r\}$. It follows from (8) that for any $x_0 \in \Gamma$ and $r > 0$,

$$(9) \quad B_\Omega(\Phi(x_0), c_\Phi r) \subset \Phi(B_\Gamma(x_0, r)) \subset B_\Omega(\Phi(x_0), C_\Phi r).$$

This, together with Lemma 3.12, yields

$$(10) \quad \mu(\Phi(B_\Gamma(x_0, r))) \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Lemma 3.13. For all $x_0 \in \Gamma$ and $r > 0$, one can find an axis-parallel rectangle R' on Ω such that

$$(11) \quad B_\Omega(\Phi(x_0), C_\Phi r) \subset R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

Furthermore, we have

$$(12) \quad \mu(R') \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)).$$

Proof. We write $(u_0, z_0) := \Phi(x_0) \in \Omega$. and define the axis-parallel rectangle

$$(13) \quad R' := \left(\prod_{i=1}^{N_0} [\max\{0, (u_0)_i - C_\Phi r\}, (u_0)_i + C_\Phi r] \right) \times \left(\prod_{j=1}^{N-N_0} [(z_0)_j - C_\Phi r, (z_0)_j + C_\Phi r] \right).$$

On the one hand, it follows immediately that $B_\Omega(\Phi(x_0), C_\Phi r) \subset R'$. On the other hand, since R' is a rectangle with half-side lengths bounded by $C_\Phi r$ in each coordinate, we have

$$R' \subset B_\Omega(\Phi(x_0), \sqrt{N} C_\Phi r).$$

The second statement follows from the first one together with Lemma 3.12. The proof of Lemma 3.13 is finished. $\square$

The following theorem demonstrates that $(\Gamma, \|\cdot\|, dw)$ is a doubling metric measure space.

Theorem 3.14. There exists a constant $C \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,

$$(14) \quad C^{-1} r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)} \leq w(B_\Gamma(x, r)) \leq C r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

In particular, there exists a constant $C_{\text{dbl}, \Gamma} \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,

$$w(B_\Gamma(x, 2r)) \leq C_{\text{dbl}, \Gamma} w(B_\Gamma(x, r)).$$

Moreover, for all $x \in \Gamma$ and $r > 0$, we have

$$(15) \quad w(B_\Gamma(x, r)) \sim w(B(x, r)).$$

Proof. The main strategy of the proof is to transfer the ball estimates from $(\Omega, \|\cdot\|, \mu)$ back to $(\Gamma, \|\cdot\|, dw)$. Fix $x \in \Gamma$ and $r > 0$. It follows from the equality (4) that

$$(16) \quad w(B_\Gamma(x, r)) = \mu(\Phi(B_\Gamma(x, r))).$$

Applying μ to the inclusions (9) and using (16) yields

$$(17) \quad \mu(B_\Omega(\Phi(x), c_\Phi r)) \leq w(B_\Gamma(x, r)) \leq \mu(B_\Omega(\Phi(x), C_\Phi r)).$$

Write $\Phi(x) = (u, z) \in \Omega$. For each $v \in R_+$ recall the linear form $L_{v,0} := \sum_{i=1}^{N_0} a_{v,i} u_i$. Combining the inequality (17) with Lemma 3.12 and the equality (6), we obtain that

$$(18) \quad w(B_\Gamma(x, r)) \simeq r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = r^N \prod_{v \in R_+} (|\langle x, v \rangle| + r)^{2k(v)}.$$

This proves (14). The second statement follows immediately from (14). The third statement follows from the inequalities (14) and (1). This completes the proof of Theorem 3.14. $\square$

4. Proof of upper bound

Lemma 4.1. Assume that $N > 1$. For $\sigma, \tau \in G$, define

$$(1) \quad \mathcal{G}_{\sigma,\tau} f(z) := \begin{cases} \frac{|f(\tau z) - f(\sigma z)|}{|\tau z - \sigma z|}, & \tau z \neq \sigma z, \\ 0, & \tau z = \sigma z. \end{cases}$$

Then for every $f \in \dot{W}^{1,N}(\mathbb{R}^N)$,

$$(2) \quad \|\mathcal{G}_{\sigma,\tau} f\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

The implicit constant is independent of $\sigma, \tau \in G$.

Proof. Let $\mathcal{M}$ be the Hardy–Littlewood maximal operator on $\mathbb{R}^N$. By [?, Theorem 3], after choosing the precise representative of f , there exists a null set $E \subset \mathbb{R}^N$ such that

$$|f(x) - f(y)| \lesssim_N |x - y|(\mathcal{M}(|\nabla f|)(x) + \mathcal{M}(|\nabla f|)(y)), \quad x, y \in \mathbb{R}^N \setminus E.$$

Since σ and τ are orthogonal, they preserve Lebesgue measure. Hence, for almost every $z \in \mathbb{R}^N$ with $\tau z \neq \sigma z$,

$$\mathcal{G}_{\sigma,\tau} f(z) \lesssim_N \mathcal{M}(|\nabla f|)(\tau z) + \mathcal{M}(|\nabla f|)(\sigma z).$$

The same estimate holds when $\tau z = \sigma z$ by the definition of $\mathcal{G}_{\sigma,\tau} f$. Therefore, using again the measure-preserving property of σ and τ and the L^N -boundedness of $\mathcal{M}$, which is valid since $N > 1$,

$$\begin{aligned} \|\mathcal{G}_{\sigma,\tau} f\|_{L^N(\mathbb{R}^N)} &\lesssim_N \|\mathcal{M}(|\nabla f|) \circ \tau\|_{L^N} + \|\mathcal{M}(|\nabla f|) \circ \sigma\|_{L^N} \\ &= 2\|\mathcal{M}(|\nabla f|)\|_{L^N} \lesssim_N \|\nabla f\|_{L^N(\mathbb{R}^N)}. \end{aligned}$$

The implicit constant is independent of $\sigma, \tau \in G$. $\square$

Lemma 4.2. Assume that $N \geq 1$, $1 < p < \infty$, and $1 \leq q \leq \infty$. For $\sigma, \tau \in G$ and $I \in \mathbb{D}_\Gamma$, let $\theta_{\sigma,\tau,I}$ be as in (??). For $f \in L^1_{\text{loc}}(\mathbb{R}^N)$, define

$$(3) \quad a_{\sigma,\tau,I}(f) := \frac{1}{|I|} \int_I (f(\tau z) - f(\sigma z)) dz.$$

We have

$$(4) \quad \left\| \{ \theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f) \}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N,\infty}} \lesssim \|f\|_{\dot{W}^{1,N}(\Gamma)}.$$

Proof. By part 1 of Lemma 8.2,

$$\|\nabla f\|_{L^N(\mathbb{R}^N)} \simeq \|f\|_{\text{osc}^{N,\infty}(\mathbb{R}^N)}.$$

For $I = \Phi^{-1}(Q) \in \mathbb{D}_\Gamma$ and $z \in I$, we use the bi-Lipschitz property (8) to see that

$$|z - z_I| \lesssim |\Phi(z) - \Phi(z_I)| \leq \frac{\sqrt{N}}{2} \ell(Q) \lesssim \ell(I).$$

Since σ and τ are orthogonal,

$$|\tau z - \sigma z| \leq 2|z - z_I| + |\tau z_I - \sigma z_I| \lesssim \ell(I) + |\tau z_I - \sigma z_I|.$$

Hence, by the definition of $\mathcal{G}_{\sigma,\tau}f$ (see (1)) and (??),

$$\begin{aligned} \theta_{\sigma,\tau,I} |a_{\sigma,\tau,I}(f)| &\leq \frac{\theta_{\sigma,\tau,I}}{|I|} \int_I |f(\tau z) - f(\sigma z)| dz \\ &\lesssim \frac{\ell(I)}{\ell(I) + |\sigma z_I - \tau z_I|} (\ell(I) + |\tau z_I - \sigma z_I|) \frac{1}{|I|} \int_I \mathcal{G}_{\sigma,\tau}f(z) dz \\ &\lesssim \ell(I) \frac{1}{|I|} \int_I \mathcal{G}_{\sigma,\tau}f(z) dz. \end{aligned}$$

Set

$$F_{\sigma,\tau}(u) := \mathcal{G}_{\sigma,\tau}f(\Phi^{-1}u)\chi_\Omega(u).$$

By the change of variables $u = \Phi(z)$, together with Lemma 4.1,

$$\|F_{\sigma,\tau}\|_{L^N(\mathbb{R}^N)} \lesssim \|\mathcal{G}_{\sigma,\tau}f\|_{L^N(\mathbb{R}^N)} \lesssim \|\nabla f\|_{L^N(\mathbb{R}^N)},$$

uniformly in $\sigma, \tau \in G$. Moreover, for $I = \Phi^{-1}(Q)$,

$$\ell(I) \frac{1}{|I|} \int_I \mathcal{G}_{\sigma,\tau}f(z) dz = \ell(Q) \frac{1}{|Q|} \int_Q F_{\sigma,\tau}(u) du = |Q|^{\frac{1}{N}-1} \int_Q F_{\sigma,\tau}(u) du.$$

By the construction of $\mathbb{D}_\Gamma$, every $I \in \mathbb{D}_\Gamma$ is of the form $I = \Phi^{-1}(Q)$ for a unique $Q \in \mathbb{D}_\Omega^0 \subset \mathbb{D}^0$. Applying Lemma 8.1 with $s = N$, we deduce that

$$\left\| \left\{ \ell(I) \frac{1}{|I|} \int_I \mathcal{G}_{\sigma,\tau}f(z) dz \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N,\infty}} \leq \left\| \left\{ |Q|^{\frac{1}{N}-1} \int_Q F_{\sigma,\tau}(u) du \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{N,\infty}} \lesssim \|F_{\sigma,\tau}\|_{L^N(\mathbb{R}^N)}.$$

Combining the preceding estimates, we deduce that

$$\left\| \{ \theta_{\sigma,\tau,I} a_{\sigma,\tau,I}(f) \}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N,\infty}} \lesssim \|\nabla f\|_{L^N(\mathbb{R}^N)} \simeq \|f\|_{\text{osc}^{N,\infty}(\mathbb{R}^N)},$$

where the final equivalence follows from Lemma 8.2, and where the implicit constant is uniform in $\sigma, \tau \in G$. $\square$

Lemma 4.3. Assume that $N \geq 1$, $1 < p < \infty$, and $1 \leq q \leq \infty$. Fix $s > 2$ and a closed Weyl chamber Γ . Let r_s be the exponent in Lemma ???. For $f \in L^1_{\text{loc}}(\mathbb{R}^N)$ and $\sigma, \tau \in G$, define

$$(5) \quad H_{\sigma, \tau}(f)(z) := f(\tau z) - f(\sigma z), \quad z \in \mathbb{R}^N.$$

Then for any $f \in OSC^{p,q}(\mathbb{R}^N)$,

$$\left\| \{\text{osc}_{r_s}(H_{\sigma, \tau}(f); I)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p,q}} \lesssim \|f\|_{OSC^{p,q}(\mathbb{R}^N)},$$

The implicit constant is independent of $\sigma, \tau \in G$.

Proof. Since σ and τ are orthogonal,

$$(H_{\sigma, \tau}(f))_{I, dx} = (f)_{\tau I, dx} - (f)_{\sigma I, dx},$$

and hence, by Minkowski's inequality,

$$\text{osc}_{r_s}(H_{\sigma, \tau}(f); I) \leq \text{osc}_{r_s}(f; \tau I) + \text{osc}_{r_s}(f; \sigma I).$$

Fix $\gamma \in \{\sigma, \tau\}$. By the construction of $\mathbb{D}_\Gamma$ and (8),

$$\gamma I \subset B(\gamma z_I, C\ell(I)), \quad |\gamma I| = |I| \simeq \ell(I)^N.$$

By Lemma 2.8, for every $I \in \mathbb{D}_\Gamma$ there exist $\nu_\gamma(I) \in \{1, \dots, 2^N\}$ and $J_I^\gamma \in \mathbb{D}^{\nu_\gamma(I)}$ such that

$$\gamma I \subset B(\gamma z_I, C\ell(I)) \subset J_I^\gamma \subset C_N B(\gamma z_I, C\ell(I)).$$

Thus, $\ell(J_I^\gamma) \simeq \ell(I)$, and $|\gamma I| \simeq |J_I^\gamma|$. This implies that

$$\text{osc}_{r_s}(f; \gamma I) \lesssim \text{osc}_{r_s}(f; J_I^\gamma).$$

Note that for each $\nu \in \{1, \dots, 2^N\}$, the map

$$I \mapsto J_I^\gamma, \quad \{I \in \mathbb{D}_\Gamma : \nu_\gamma(I) = \nu\} \longrightarrow \mathbb{D}^\nu,$$

has uniformly bounded multiplicity. That is,

$$\sup_{J \in \mathbb{D}^\nu} \#\{I \in \mathbb{D}_\Gamma : \nu_\gamma(I) = \nu, J_I^\gamma = J\} \lesssim 1.$$

Therefore, by Lemma ??,

$$\begin{aligned} \left\| \{\text{osc}_{r_s}(f; \gamma I)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p,q}} &\lesssim \sum_{\nu=1}^{2^N} \left\| \left\{ \text{osc}_{r_s}(f; J_I^\gamma) \right\}_{\substack{I \in \mathbb{D}_\Gamma \\ \nu_\gamma(I) = \nu}} \right\|_{\ell^{p,q}} \\ &\lesssim \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_{r_s}(f; J)\}_{J \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}} = \|f\|_{OSC^{p,q}(\mathbb{R}^N)}. \end{aligned}$$

Applying this estimate with $\gamma = \sigma$ and $\gamma = \tau$, we obtain

$$\left\| \{\text{osc}_{r_s}(H_{\sigma, \tau}(f); I)\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p,q}} \lesssim \|f\|_{OSC^{p,q}(\mathbb{R}^N)}.$$

This completes the proof of Lemma 4.3. $\square$

5. Proof of lower bound

Definition 5.1. Let $1 \leq r < \infty$. For a measurable set $E \subset \mathbb{R}^N$ with $0 < w(E) < \infty$, set

$$(f)_{E,dw} := \frac{1}{w(E)} \int_E f dw$$

and

$$\text{osc}_{r,dw}(f; E) := \left(\frac{1}{w(E)} \int_E |f - (f)_{E,dw}|^r dw \right)^{1/r}.$$

For $0 < p < \infty$ and $0 < q \leq \infty$, define

$$\|f\|_{OS C_{dw}^{p,q}} := \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_{1,dw}(f; Q)\}_{Q \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}}.$$

Lemma 5.2. Let $0 < p < \infty$, $0 < q \leq \infty$, and $1 \leq r < \infty$. For $f \in L_{\text{loc}}^r(\mathbb{R}^N, dx)$, set

$$\text{osc}_{r,dw}(f; Q) := \left(\frac{1}{w(Q)} \int_Q |f - (f)_{Q,dw}|^r dw \right)^{1/r}$$

and

$$\|f\|_{OS C_{dw}^{p,q}} := \sum_{\nu=1}^{2^N} \left\| \{\text{osc}_{r,dw}(f; Q)\}_{Q \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}}.$$

Then

$$\|f\|_{OS C^{p,q}(\mathbb{R}^N)} \sim \|f\|_{OS C_{dw}^{p,q}},$$

where the implicit constants are independent of f .

Proof. By Lemma 2.1,

$$dw = \varpi dx, \quad \varpi \in A_\infty(\mathbb{R}^N, dx).$$

Moreover, both dx and dw are doubling measures. Hence dx and dw satisfy the A_∞ -relation in the sense of [?, Proposition 1.2]. Therefore,

$$\|f\|_{\text{Osc}^{p,q}(dx)} \sim \|f\|_{\text{Osc}^{p,q}(dw)}$$

by [?, Proposition 1.2].

By [?, (3.1) and Lemma 3.4], together with Lemma 2.8, these two oscillatory norms are equivalent, respectively, to the dyadic norms

$$\|f\|_{OS C^{p,q}(\mathbb{R}^N)} \quad \text{and} \quad \|f\|_{OS C_{dw}^{p,q}}.$$

The assertion follows. $\square$

Lemma 5.3. Fix $j \in \{1, \dots, N\}$. For every $Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}^\nu$, let

$$r_Q := \sqrt{N} \ell(Q), \quad B_Q := B(c_Q, r_Q),$$

and define

$$\widehat{B}_Q := B(c_Q + 5r_Q e_j, r_Q).$$

Then

$$(1) \quad w(Q) \sim w(B_Q) \sim w(\widehat{B}_Q),$$

and there is a constant C_N , independent of Q , such that

$$Q \cup \widehat{B}_Q \subset C_N Q.$$

Moreover, for every $(x, y) \in Q \times \widehat{B}_Q$, the kernel $K_j(x, y)$ does not change sign and

$$(2) \quad |K_j(x, y)| \gtrsim \frac{1}{w(Q)}.$$

Proof. Since Q is a Euclidean cube,

$$B\left(c_Q, \frac{\ell(Q)}{2}\right) \subset Q \subset B_Q.$$

Hence the doubling property (2) gives

$$w(B_Q) \lesssim w\left(B\left(c_Q, \frac{\ell(Q)}{2}\right)\right) \leq w(Q) \leq w(B_Q),$$

and therefore

$$w(Q) \sim w(B_Q).$$

Since the distance between the centers of B_Q and $\widehat{B}_Q$ is $5r_Q$,

$$\widehat{B}_Q \subset 6B_Q.$$

Similarly,

$$B_Q \subset B(c_Q + 5r_Q e_j, 6r_Q) = 6\widehat{B}_Q.$$

The doubling property therefore yields

$$w(B_Q) \sim w(\widehat{B}_Q),$$

which proves (1).

For $y \in \widehat{B}_Q$,

$$\|y - c_Q\| \leq 6r_Q = 6\sqrt{N}\ell(Q).$$

Thus $Q \cup \widehat{B}_Q$ is contained in a fixed concentric enlargement $C_N Q$.

Finally, $Q \subset B_Q$, so Lemma 2.4, applied to B_Q , shows that K_j has a fixed sign on $Q \times \widehat{B}_Q$ and

$$|K_j(x, y)| \gtrsim \frac{1}{w(B_Q)} \sim \frac{1}{w(Q)}.$$

□

Lemma 5.4. Let $f \in L^1_{\text{loc}}(\mathbb{R}^N, dw)$ be real-valued, and assume that $[R_{\text{Dunkl}, j}, M_f]$ is a compact operator on $L^2(\mathbb{R}^N, dw)$. For every $\nu \in \{1, \dots, 2^N\}$, there exist NWO sequences

$$\{e_{Q,m}\}_{Q \in \mathbb{D}^\nu}, \quad \{h_{Q,m}\}_{Q \in \mathbb{D}^\nu}, \quad m = 1, 2,$$

such that, for every $Q \in \mathbb{D}^\nu$,

$$\text{osc}_{1,dw}(f; Q) \lesssim \sum_{m=1}^2 \left| \langle [R_{\text{Dunkl}, j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right|.$$

Proof. Fix $\nu \in \{1, \dots, 2^N\}$ and $Q \in \mathbb{D}^\nu$. Set

$$r_Q := \sqrt{N}\ell(Q), \quad B_Q := B(c_Q, r_Q).$$

Since

$$B(c_Q, \ell(Q)/2) \subset Q \subset B_Q,$$

by (2),

$$w(B_Q) \simeq w(Q).$$

By Lemma 2.4, for

$$\widehat{B}_Q := B(c_Q + 5r_Q e_j, r_Q),$$

the kernel K_j has a fixed sign on $B_Q \times \widehat{B}_Q$ and

$$|K_j(x, y)| \gtrsim \frac{1}{w(Q)}, \quad (x, y) \in Q \times \widehat{B}_Q.$$

Moreover,

$$\widehat{B}_Q \subset 6B_Q, \quad B_Q \subset 6\widehat{B}_Q,$$

so (2) implies

$$w(\widehat{B}_Q) \simeq w(Q).$$

Since $\widehat{B}_Q \subset B(c_Q, 6r_Q)$, there is a fixed $C_N \geq 1$ such that

$$\widehat{B}_Q \subset C_N Q, \quad w(C_N Q) \simeq w(Q).$$

Let α_Q be a dw -median of f on $\widehat{B}_Q$, and define

$$\begin{aligned} E_{Q,1} &:= \{x \in Q : f(x) < \alpha_Q\}, & F_{Q,1} &:= \{y \in \widehat{B}_Q : f(y) \geq \alpha_Q\}, \\ E_{Q,2} &:= \{x \in Q : f(x) \geq \alpha_Q\}, & F_{Q,2} &:= \{y \in \widehat{B}_Q : f(y) \leq \alpha_Q\}. \end{aligned}$$

Then

$$w(F_{Q,m}) \geq \frac{1}{2} w(\widehat{B}_Q) \simeq w(Q), \quad m = 1, 2.$$

For each $m = 1, 2$, $f(y) - f(x)$ has a fixed sign on $E_{Q,m} \times F_{Q,m}$, and

$$|f(x) - \alpha_Q| \leq |f(y) - f(x)|, \quad (x, y) \in E_{Q,m} \times F_{Q,m}.$$

Since K_j also has a fixed sign on $Q \times \widehat{B}_Q$, we obtain

$$\begin{aligned} \frac{1}{w(Q)} \int_{E_{Q,m}} |f - \alpha_Q| dw &\leq \frac{1}{w(Q)w(F_{Q,m})} \int_{E_{Q,m}} \int_{F_{Q,m}} |f(y) - f(x)| dw(y) dw(x) \\ &\lesssim \frac{1}{w(Q)} \left| \int_{E_{Q,m}} \int_{F_{Q,m}} (f(y) - f(x)) K_j(x, y) dw(y) dw(x) \right|. \end{aligned}$$

Define

$$e_{Q,m} := w(C_N Q)^{-1/2} \chi_{F_{Q,m}}, \quad h_{Q,m} := w(Q)^{-1/2} \chi_{E_{Q,m}}.$$

By the off-diagonal kernel identity

$$K_{[R_{\text{Dunkl},j}, M_f]}(x, y) = (f(y) - f(x)) K_j(x, y),$$

and $w(C_N Q) \simeq w(Q)$,

$$\frac{1}{w(Q)} \int_{E_{Q,m}} |f - \alpha_Q| dw \lesssim \left| \langle [R_{\text{Dunkl},j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right|.$$

Furthermore,

$$\text{supp } h_{Q,m} \subset Q, \quad \text{supp } e_{Q,m} \subset C_N Q.$$

For any fixed $s > 2$,

$$\begin{aligned} \|h_{Q,m}\|_{L^s(\mathbb{R}^N, dw)} &\leq w(Q)^{1/s-1/2}, \\ \|e_{Q,m}\|_{L^s(\mathbb{R}^N, dw)} &\leq w(C_N Q)^{1/s-1/2} \leq w(Q)^{1/s-1/2}. \end{aligned}$$

Hence, $\{e_{Q,m}\}_{Q \in \mathbb{D}^N}$ and $\{h_{Q,m}\}_{Q \in \mathbb{D}^N}$ are NWO sequences for $m = 1, 2$.

Finally,

$$|(f)_{Q,dw} - \alpha_Q| \leq \frac{1}{w(Q)} \int_Q |f - \alpha_Q| dw,$$

and therefore

$$\begin{aligned} \text{osc}_{1,dw}(f; Q) &\leq \frac{2}{w(Q)} \int_Q |f - \alpha_Q| dw \\ &= 2 \sum_{m=1}^2 \frac{1}{w(Q)} \int_{E_{Q,m}} |f - \alpha_Q| dw \\ &\lesssim \sum_{m=1}^2 \left| \langle [R_{\text{Dunkl},j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right|. \end{aligned}$$

This completes the proof of Lemma 5.4. $\square$

Theorem 5.5. Assume that $N \geq 1$, $1 < p < \infty$, and $1 \leq q \leq \infty$. Fix $j \in \{1, \dots, N\}$. Let

$$f \in L^1_{\text{loc}}(\mathbb{R}^N, dx).$$

If

$$[R_{\text{Dunkl},j}, M_f] \in S^{p,q}(L^2(\mathbb{R}^N, dw)),$$

then $f \in OSC^{p,q}(\mathbb{R}^N)$ and

$$(3) \quad \|f\|_{OSC^{p,q}(\mathbb{R}^N)} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}.$$

Proof. Assume first that f is real valued. Fix $\nu \in \{1, \dots, 2^N\}$. By Lemma 5.4, for $m = 1, 2$ there are NWO sequences

$$\{e_{Q,m}\}_{Q \in \mathbb{D}^\nu}, \quad \{h_{Q,m}\}_{Q \in \mathbb{D}^\nu},$$

such that

$$\text{osc}_{1,dw}(f; Q) \lesssim \sum_{m=1}^2 \left| \langle [R_{\text{Dunkl},j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right|.$$

By the quasi triangle inequality in $\ell^{p,q}$ and Lemma 2.9,

$$\begin{aligned} \left\| \{\text{osc}_{1,dw}(f; Q)\}_{Q \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}} &\lesssim \sum_{m=1}^2 \left\| \left\{ \langle [R_{\text{Dunkl},j}, M_f] e_{Q,m}, h_{Q,m} \rangle_{L^2(\mathbb{R}^N, dw)} \right\}_{Q \in \mathbb{D}^\nu} \right\|_{\ell^{p,q}} \\ &\lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}. \end{aligned}$$

Summing over ν and using Definition 5.1, we obtain

$$\|f\|_{OSC^{p,q}_{dw}} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}.$$

By Lemma 5.2 with $r = 1$,

$$(4) \quad \|f\|_{OSC^{p,q}(\mathbb{R}^N)} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}.$$

For general f , let C denote complex conjugation on $L^2(\mathbb{R}^N, dw)$. Since K_j is real valued by its heat kernel representation,

$$CR_{\text{Dunkl},j}C = R_{\text{Dunkl},j}, \quad C[R_{\text{Dunkl},j}, M_f]C = [R_{\text{Dunkl},j}, M_{\bar{f}}].$$

Hence,

$$\begin{aligned} [R_{\text{Dunkl},j}, M_{\text{Re } f}] &= \frac{1}{2}([R_{\text{Dunkl},j}, M_f] + C[R_{\text{Dunkl},j}, M_f]C), \\ [R_{\text{Dunkl},j}, M_{\text{Im } f}] &= \frac{1}{2i}([R_{\text{Dunkl},j}, M_f] - C[R_{\text{Dunkl},j}, M_f]C). \end{aligned}$$

Since conjugation preserves singular values,

$$\|[R_{\text{Dunkl},j}, M_{\text{Re } f}]\|_{S^{p,q}} + \|[R_{\text{Dunkl},j}, M_{\text{Im } f}]\|_{S^{p,q}} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}}.$$

By (4) applied to $\text{Re } f$ and $\text{Im } f$,

$$\|\text{Re } f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)} + \|\text{Im } f\|_{\text{OSC}^{p,q}(\mathbb{R}^N)} \lesssim \|[R_{\text{Dunkl},j}, M_f]\|_{S^{p,q}(L^2(\mathbb{R}^N, dw))}.$$

Since

$$\text{osc}_1(f; Q) \leq \text{osc}_1(\text{Re } f; Q) + \text{osc}_1(\text{Im } f; Q),$$

the quasi triangle inequality in $\ell^{p,q}$ proves (3). This completes the proof of Theorem 5.5. $\square$

6. Spectral asymptotic formula

Definition 6.1. For $g \in G$ and $j = 1, \dots, N$, set

$$d_{g,j} := g^{-1}e_j, \quad f_g(x) := f(gx).$$

For $\alpha \in R_+$, define $S_\alpha : \mathbb{C}^G \rightarrow \mathbb{C}^G$ by

$$(S_\alpha z)_g := z_{g\sigma_\alpha}, \quad g \in G.$$

For $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, $x \in \Gamma_+$, and $\theta \in \mathbb{S}^{N-1}$, define

$$\begin{aligned} c_{f,j}(x, \theta) &:= \text{diag}_{g \in G} \left(\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\ &\quad + \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha. \end{aligned}$$

For smooth f , the quotients in the second term are understood by their smooth extensions across the corresponding reflection hyperplanes. For $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, $c_{f,j}$ is understood almost everywhere on $\Gamma \times \mathbb{S}^{N-1}$.

Lemma 6.2. Let $f \in W_{\text{loc}}^{1,1}(\mathbb{R}^N)$. For every $u \in C_c^\infty(\mathbb{R}^N)$,

$$(1) \quad [T_{e_j}, M_f]u(x) = \partial_j f(x)u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{f(x) - f(\sigma_\alpha x)}{\langle \alpha, x \rangle} u(\sigma_\alpha x)$$

for almost every $x \in \mathbb{R}^N$. If $N > 1$ and $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, then, for every $\alpha \in R_+$,

$$(2) \quad \left\| \frac{f - f \circ \sigma_\alpha}{\langle \alpha, \cdot \rangle} \right\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

Proof. Pairing α and $-\alpha$ in the definition of T_{e_j} , and using $k(-\alpha) = k(\alpha)$ and $\sigma_{-\alpha} = \sigma_\alpha$, we have

$$T_{e_j}u(x) = \partial_j u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{u(x) - u(\sigma_\alpha x)}{\langle \alpha, x \rangle}.$$

Therefore,

$$[T_{e_j}, M_f]u(x) = \partial_j f(x)u(x) + \sum_{\alpha \in R_+} k(\alpha) \langle \alpha, e_j \rangle \frac{f(x) - f(\sigma_\alpha x)}{\langle \alpha, x \rangle} u(\sigma_\alpha x),$$

which proves (1).

Since R is normalized,

$$x - \sigma_\alpha x = \langle \alpha, x \rangle \alpha, \quad \|x - \sigma_\alpha x\| = \sqrt{2} |\langle \alpha, x \rangle|.$$

Hence, for almost every $x \notin \alpha^\perp$,

$$\frac{|f(x) - f(\sigma_\alpha x)|}{|\langle \alpha, x \rangle|} = \sqrt{2} \mathcal{G}_{I, \sigma_\alpha} f(x).$$

By Lemma 4.1 (2),

$$\left\| \frac{f - f \circ \sigma_\alpha}{\langle \alpha, \cdot \rangle} \right\|_{L^N(\mathbb{R}^N, dx)} \lesssim_{N, G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

This completes the proof of Lemma 6.2. $\square$

Lemma 6.3. Let Γ_+ be an open Weyl chamber, let Γ be its closure, and let R_+ be the corresponding positive subsystem. Set

$$\rho(x) := \varpi(x)^{1/2},$$

and define

$$\begin{aligned} U : L^2(\mathbb{R}^N, dw) &\longrightarrow L^2(\Gamma, dw; \mathbb{C}^G), & (Uu)_g(x) &:= u(gx), \\ V : L^2(\Gamma, dw; \mathbb{C}^G) &\longrightarrow L^2(\Gamma, dx; \mathbb{C}^G), & (Vh)(x) &:= \rho(x)h(x). \end{aligned}$$

Then U and V are unitary. If

$$\tilde{T}_j := VUT_{e_j}U^{-1}V^{-1}, \quad \tilde{P} := VU(-\Delta_{\text{Dunkl}})U^{-1}V^{-1},$$

then, on $C_c^\infty(\Gamma_+; \mathbb{C}^G)$,

$$(3) \quad (\tilde{T}_j h)_g = \partial_{d_{g,j}} h_g - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} h_{g\sigma_\alpha},$$

$$(4) \quad \tilde{P} = -\Delta_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Assume in addition that $N > 1$, and let

$$\tilde{Q} := VU(-\Delta_{\text{Dunkl}})^{-1/2}U^{-1}V^{-1},$$

where $(-\Delta_{\text{Dunkl}})^{-1/2}$ is defined by the spectral calculus. Then, for every $\chi \in C_c^\infty(\Gamma_+)$, the operator $M_\chi \tilde{Q} M_\chi$, initially defined on its natural domain, admits a bounded extension to $L^2(\Gamma, dx; \mathbb{C}^G)$, and this extension is a classical matrix-valued pseudodifferential operator of order -1 . Its principal symbol is

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

Consequently, if $x \in \Gamma_+$, $\xi \neq 0$, and $\chi \in C_c^\infty(\Gamma_+)$ is identically 1 in a neighborhood of x , then

$$(5) \quad \sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}.$$

Proof. The Weyl chambers $g\Gamma_+$, $g \in G$, are pairwise disjoint and cover $\mathbb{R}^N$ up to the reflection hyperplanes; see [5, Theorem 2.12]. Since dw is G -invariant,

$$\begin{aligned} \|Uu\|_{L^2(\Gamma, dw; \mathbb{C}^G)}^2 &= \sum_{g \in G} \int_{\Gamma} |u(gx)|^2 dw(x) \\ &= \int_{\mathbb{R}^N} |u(x)|^2 dw(x). \end{aligned}$$

Conversely, for $h = (h_g)_{g \in G} \in L^2(\Gamma, dw; \mathbb{C}^G)$, define $u(gx) := h_g(x)$ for $x \in \Gamma_+$. This determines u almost everywhere on $\mathbb{R}^N$ and gives $Uu = h$. Hence U is unitary. Since $dw = \rho^2 dx$ and $\rho > 0$ on Γ_+ , V is unitary as well.

Pairing the roots α and $-\alpha$ in the definition of T_{e_j} , changing variables in the root system by g , and using the G -invariance of k , we obtain, for $h \in C_c^\infty(\Gamma_+; \mathbb{C}^G)$,

$$(UT_{e_j}U^{-1}h)_g = \partial_{d_{g,j}}h_g + \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (h_g - h_{g\sigma_\alpha}).$$

On Γ_+ ,

$$\nabla \log \rho(x) = \sum_{\alpha \in R_+} k(\alpha) \frac{\alpha}{\langle \alpha, x \rangle},$$

and therefore

$$\rho \partial_{d_{g,j}} \rho^{-1} = \partial_{d_{g,j}} - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle}.$$

Conjugation by V cancels the diagonal terms and gives (3).

Similarly, pairing α and $-\alpha$ in the formula for the Dunkl Laplacian and using $\|\alpha\|^2 = 2$, we have

$$\begin{aligned} (U(-\Delta_{\text{Dunkl}})U^{-1}h)_g &= -\Delta h_g - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{\partial_\alpha h_g}{\langle \alpha, x \rangle} \\ &\quad + 2 \sum_{\alpha \in R_+} k(\alpha) \frac{h_g - h_{g\sigma_\alpha}}{\langle \alpha, x \rangle^2}. \end{aligned}$$

Writing $\phi = \log \rho$,

$$\rho(-\Delta - 2\nabla\phi \cdot \nabla)\rho^{-1} = -\Delta + |\nabla\phi|^2 + \Delta\phi,$$

while

$$\Delta\phi = - \sum_{\alpha \in R_+} k(\alpha) \frac{\|\alpha\|^2}{\langle \alpha, x \rangle^2} = -2 \sum_{\alpha \in R_+} \frac{k(\alpha)}{\langle \alpha, x \rangle^2}.$$

The latter term cancels the diagonal part of

$$2 \sum_{\alpha \in R_+} k(\alpha) \frac{I_{\mathbb{C}^G} - S_\alpha}{\langle \alpha, x \rangle^2},$$

and hence

$$\tilde{P} = -\Delta I_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2},$$

which proves (4).

Assume now that $N > 1$, fix $\chi \in C_c^\infty(\Gamma_+)$, and set

$$K := \text{supp } \chi.$$

On Γ_+ , write

$$\tilde{P} = -\Delta I_{\mathbb{C}^G} + W(x), \quad W(x) = |\nabla \log \rho(x)|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Thus W is a smooth Hermitian matrix-valued function on Γ_+ .

Choose open sets

$$K \Subset \Omega_0 \Subset \Omega_1 \Subset \Gamma_+$$

and $\eta \in C_c^\infty(\Gamma_+)$ such that

$$0 \leq \eta \leq 1, \quad \eta = 1 \quad \text{on a neighborhood of } \overline{\Omega_1}.$$

Choose a flat torus M containing $\text{supp } \eta$ in one Euclidean coordinate chart. Since η is compactly supported in that chart, the differential operator $M_\eta \widetilde{P} M_\eta$ extends smoothly by zero to M . Define on $L^2(M; \mathbb{C}^G)$

$$(6) \quad P_0 := M_\eta \widetilde{P} M_\eta + M_{1-\eta}(-\Delta_M + 1)M_{1-\eta}.$$

It is a nonnegative self-adjoint elliptic differential operator of order 2. Indeed,

$$\langle P_0 u, u \rangle = \langle \widetilde{P}(\eta u), \eta u \rangle + \|\nabla((1-\eta)u)\|_2^2 + \|(1-\eta)u\|_2^2 \geq 0,$$

and its principal symbol is

$$p_{0,2}(x, \xi) = (\eta(x)^2 + (1-\eta(x))^2)|\xi|^2 I_{\mathbb{C}^G},$$

which is elliptic since

$$\eta^2 + (1-\eta)^2 \geq \frac{1}{2}.$$

Moreover,

$$(7) \quad P_0 = \widetilde{P} \quad \text{on } \Omega_1.$$

Let

$$H_t := e^{-t\widetilde{P}}, \quad H_t^0 := e^{-tP_0}.$$

We first consider

$$B_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} H_t^0 dt.$$

Let Π_0 be the orthogonal projection onto $\ker P_0$. Since P_0 is elliptic on the closed manifold M , Π_0 is smoothing. Applying the complex powers calculus [?, Theorem 2 and p. 290] to $P_0 + \Pi_0$, we obtain

$$(P_0 + \Pi_0)^{-1/2} \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G)$$

with principal symbol $p_{0,2}^{-1/2}$.

Set

$$F(\lambda) := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} e^{-t\lambda} dt, \quad \lambda \geq 0.$$

Then $B_0 = F(P_0)$, and for $\lambda > 0$,

$$F(\lambda) - \lambda^{-1/2} = -\frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} e^{-t\lambda} dt.$$

Hence

$$F(\lambda) - \lambda^{-1/2} = O(\lambda^{-m}) \quad (\lambda \rightarrow \infty)$$

for every $m > 0$. On $\ker P_0$ the difference between B_0 and $(P_0 + \Pi_0)^{-1/2}$ is finite rank. It follows from elliptic spectral regularity that

$$B_0 - (P_0 + \Pi_0)^{-1/2} \in \Psi^{-\infty}(M; \mathbb{C}^G).$$

Consequently,

$$(8) \quad B_0 \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G), \quad \sigma_{-1}(B_0)(x, \xi) = p_{0,2}(x, \xi)^{-1/2}.$$

In particular, by (7),

$$(9) \quad \sigma_{-1}(B_0)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}, \quad x \in \Omega_1.$$

We next compare H_t and H_t^0 in the interior. Choose $\zeta \in C_c^\infty(\Omega_1)$ such that

$$\zeta = 1 \quad \text{on a neighborhood of } K.$$

Since $P_0 = \widetilde{P}$ on Ω_1 , Duhamel's formula gives

$$(10) \quad M_\chi(H_t - H_t^0)M_\chi = - \int_0^t M_\chi H_{t-s}[\widetilde{P}, M_\zeta]H_s^0 M_\chi ds.$$

Indeed, for data supported in K ,

$$v(t) := M_\zeta H_t^0 M_\chi u$$

satisfies

$$(\partial_t + \widetilde{P})v = [\widetilde{P}, M_\zeta]H_t^0 M_\chi u, \quad v(0) = M_\chi u,$$

which yields (10).

Let $h_t(x, y)$ be the Dunkl heat kernel. The integral kernel of H_t , with respect to Lebesgue measure on Γ , is the matrix

$$(11) \quad \widetilde{h}_t(x, y)_{g, \ell} = \rho(x)\rho(y)h_t(gx, \ell y), \quad g, \ell \in G.$$

For $x, y \in \Gamma$,

$$d(gx, \ell y) = d(x, y) = |x - y|.$$

Together with the mixed derivative estimate for the Dunkl heat kernel [?, Theorem 3.1], this implies that, whenever $E, F \Subset \Gamma_+$ and $\text{dist}(E, F) > 0$, for every pair of multi-indices α, β and every $m > 0$,

$$(12) \quad \sup_{\substack{x \in E \\ y \in F}} \left\| \partial_x^\alpha \partial_y^\beta \widetilde{h}_t(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

The heat kernel of the smooth elliptic operator P_0 satisfies the same off-diagonal estimate on M .

Since

$$\text{supp}[\widetilde{P}, M_\zeta] \cap K = \emptyset,$$

the two heat kernels occurring in (10) are evaluated between compact sets separated by a positive distance. The standard Gaussian bounds underlying (12) therefore give, for every multi-indices α, β and every $m > 0$,

$$(13) \quad \sup_{x, y} \left\| \partial_x^\alpha \partial_y^\beta K_{M_\chi(H_t - H_t^0)M_\chi}(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

Hence

$$(14) \quad S_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} M_\chi(H_t - H_t^0)M_\chi dt \in \Psi^{-\infty}.$$

It remains to treat the large-time part. Let $K_1 \Subset \Gamma_+$ be a compact set containing K . By (11), the heat-kernel derivative estimate cited above, and (2), for $x, y \in K_1$ and $t \geq 1$,

$$V(gx, \ell y, \sqrt{t})^{-1} \lesssim_{K_1} t^{-N/2}.$$

Since ρ and all its derivatives are bounded on K_1 , it follows that, for every pair of multi-indices α, β ,

$$(15) \quad \sup_{x, y \in K_1} \left\| \partial_x^\alpha \partial_y^\beta \widetilde{h}_t(x, y) \right\| \lesssim_{\alpha, \beta, K_1} t^{-N/2}, \quad t \geq 1.$$

Since $N > 1$,

$$\int_1^\infty t^{-1/2} t^{-N/2} dt < \infty.$$

Thus all derivatives of

$$K_{S_\infty}(x, y) := \frac{\chi(x)\chi(y)}{\sqrt{\pi}} \int_1^\infty t^{-1/2} \widetilde{h}_t(x, y) dt$$

converge locally uniformly, and therefore

$$(16) \quad S_\infty := \frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} M_\chi H_t M_\chi dt \in \Psi^{-\infty}.$$

For completeness, the preceding integrals give the canonical bounded localization of $\widetilde{P}^{-1/2}$. Indeed, for $\varepsilon > 0$,

$$(\widetilde{P} + \varepsilon)^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} e^{-\varepsilon t} H_t dt$$

by the spectral theorem. The estimates (13) and (15) imply that

$$M_\chi (\widetilde{P} + \varepsilon)^{-1/2} M_\chi$$

converges in operator norm, as $\varepsilon \downarrow 0$, to

$$\frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} M_\chi H_t M_\chi dt.$$

On the natural domain of $\widetilde{P}^{-1/2}$, this limit agrees with $M_\chi \widetilde{P}^{-1/2} M_\chi$; hence it is its bounded extension.

Combining (14), (16), and the definition of B_0 , we obtain

$$M_\chi \widetilde{Q} M_\chi = M_\chi B_0 M_\chi + S_0 + S_\infty.$$

By (8), $M_\chi B_0 M_\chi$ is a classical matrix-valued pseudodifferential operator of order -1 , while $S_0 + S_\infty$ is smoothing. Therefore

$$M_\chi \widetilde{Q} M_\chi \in \Psi_{\text{cl}}^{-1},$$

and, by (9),

$$\sigma_{-1}(M_\chi \widetilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

If $\chi = 1$ in a neighborhood of x , this reduces to

$$\sigma_{-1}(M_\chi \widetilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G},$$

which proves (5). □

Lemma 6.4. Let $j \in \{1, \dots, N\}$ and $f \in C_c^\infty(\mathbb{R}^N)$, and set

$$A_{f,j} := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

Then, for every $\chi \in C_c^\infty(\Gamma_+)$,

$$M_\chi A_{f,j} M_\chi$$

is a compactly supported classical matrix valued pseudodifferential operator of order -1 , with principal symbol

$$(17) \quad \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) = \frac{\chi(x)^2}{\|\xi\|} c_{f,j}\left(x, \frac{\xi}{\|\xi\|}\right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

Proof. Set

$$F(x) := \text{diag}_{g \in G} (f_g(x)).$$

Since $VUM_fU^{-1}V^{-1} = M_F$ and $R_{\text{Dunkl},j} = T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$,

$$(18) \quad A_{f,j} = [\widetilde{T}_j, M_F]\widetilde{Q} + \widetilde{T}_j[\widetilde{Q}, M_F].$$

By (3),

$$(19) \quad \begin{aligned} [\widetilde{T}_j, M_F] &= \text{diag}_{g \in G} (\langle d_{g,j}, \nabla f_g \rangle) \\ &+ \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g - f_{g\sigma_\alpha}) \right) S_\alpha. \end{aligned}$$

Fix $\chi \in C_c^\infty(\Gamma_+)$. Choose $\eta, \psi \in C_c^\infty(\Gamma_+)$ such that $\eta = 1$ on a neighborhood of $\text{supp } \chi$ and $\psi = 1$ on a neighborhood of $\text{supp } \eta$, and set

$$Q_\psi := M_\psi \widetilde{Q} M_\psi.$$

By Lemma 6.3, Q_ψ is a classical matrix valued pseudodifferential operator of order -1 , and on a neighborhood of $\text{supp } \eta$,

$$(20) \quad q_{-1}(x, \xi) = \|\xi\|^{-1} I_{\mathbb{C}^G}.$$

We use the Kohn–Nirenberg convention

$$\text{Op}(a)u(x) = (2\pi)^{-N} \int_{\mathbb{R}^{2N}} e^{i(x-y) \cdot \xi} a(x, \xi) u(y) dy d\xi.$$

If a and b are classical symbols, the composition formula [?, Theorem 18.1.8, p. 71] gives

$$(21) \quad a \# b \sim \sum_{\beta \in \mathbb{N}_0^N} \frac{1}{\beta!} \partial_\xi^\beta a D_x^\beta b, \quad D_x := \frac{1}{i} \partial_x.$$

Write, on a neighborhood of $\text{supp } \eta$,

$$\sigma(Q_\psi)(x, \xi) \sim q_{-1}(x, \xi) + q_{-2}(x, \xi) + \cdots.$$

By (4),

$$\sigma(\widetilde{P})(x, \xi) \sim p_2(x, \xi) + p_1(x, \xi) + p_0(x, \xi),$$

where

$$p_2(x, \xi) = \|\xi\|^2 I_{\mathbb{C}^G}, \quad p_1(x, \xi) = 0.$$

Since $\widetilde{Q} = \widetilde{P}^{-1/2}$, $\widetilde{Q}\widetilde{P}\widetilde{Q} = I$. Applying (21) on the region where $\psi = 1$ and comparing the homogeneous terms of degree -1 , the terms containing x -derivatives of p_2 and q_{-1} vanish, and hence

$$q_{-2}p_2q_{-1} + q_{-1}p_2q_{-2} = 0.$$

Together with (20), this gives

$$(22) \quad q_{-2}(x, \xi) = 0$$

on a neighborhood of $\text{supp } \eta$.

Applying (21) to the commutator $[Q_\psi, M_F]$, and using (20)–(22), we obtain, on the same neighborhood,

$$\begin{aligned} \sigma_{-2}([Q_\psi, M_F])(x, \xi) &= \frac{1}{i} \sum_{\ell=1}^N \partial_{\xi_\ell} (\|\xi\|^{-1}) \partial_{x_\ell} F(x) \\ (23) \qquad \qquad \qquad &= i \|\xi\|^{-3} \operatorname{diag}_{g \in G} (\langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Since $\psi = 1$ on a neighborhood of $\operatorname{supp} \eta$ and $\eta = 1$ on a neighborhood of $\operatorname{supp} \chi$,

$$\begin{aligned} M_\chi[\tilde{T}_j, M_F] \tilde{Q} M_\chi &= M_\chi[\tilde{T}_j, M_F] Q_\psi M_\chi, \\ M_\chi \tilde{T}_j[\tilde{Q}, M_F] M_\chi &= M_\chi \tilde{T}_j M_\eta [Q_\psi, M_F] M_\chi. \end{aligned}$$

Both operators on the right are classical matrix valued pseudodifferential operators of order -1 .

By (19), (20), and the order-zero part of the composition formula,

$$\begin{aligned} \sigma_{-1}(M_\chi[\tilde{T}_j, M_F] \tilde{Q} M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \operatorname{diag}_{g \in G} (\langle d_{g,j}, \nabla f_g(x) \rangle) \\ (24) \qquad \qquad \qquad &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \operatorname{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha. \end{aligned}$$

Under the above Kohn–Nirenberg convention, (3) gives

$$\sigma_1(\tilde{T}_j)(x, \xi) = i \operatorname{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle).$$

Therefore, by (23),

$$\begin{aligned} \sigma_{-1}(M_\chi \tilde{T}_j[\tilde{Q}, M_F] M_\chi)(x, \xi) \\ (25) \qquad \qquad \qquad &= -\chi(x)^2 \|\xi\|^{-3} \operatorname{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle \langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Combining (24) and (25), and setting

$$\theta := \frac{\xi}{\|\xi\|},$$

we obtain

$$\begin{aligned} \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \operatorname{diag}_{g \in G} \left(\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\ &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \operatorname{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha \\ &= \frac{\chi(x)^2}{\|\xi\|} \mathfrak{c}_{f,j}(x, \theta). \end{aligned}$$

This proves (17). □

Lemma 6.5. Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Then

$$\left(\int_\Gamma \int_{\mathbb{S}^{N-1}} \operatorname{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

Proof. By the definition of $\mathfrak{c}_{f,j}$, the unitary invariance of the Schatten norm, and $\|d_{g,j}\| = 1$,

$$\begin{aligned} \|\mathfrak{c}_{f,j}(x, \theta)\|_{S^N} &\lesssim_{R,k} \left(\sum_{g \in G} |\nabla f_g(x)|^N \right)^{1/N} \\ &\quad + \sum_{\alpha \in R_+} \left(\sum_{g \in G} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N \right)^{1/N}. \end{aligned}$$

The chamber decomposition used in Lemma 6.3 and the orthogonality of $g \in G$ imply

$$\sum_{g \in G} \int_{\Gamma} |\nabla f_g(x)|^N dx = \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N.$$

Moreover, since $\|\alpha\|^2 = 2$,

$$|x - \sigma_\alpha x| = \sqrt{2} |\langle \alpha, x \rangle|,$$

and hence

$$\left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right| = \sqrt{2} \mathcal{G}_{g, g\sigma_\alpha} f(x)$$

for almost every $x \in \mathbb{R}^N$. By Lemma 4.1,

$$\int_{\Gamma} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N dx \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N$$

uniformly in $g \in G$ and $\alpha \in R_+$. Since G and R_+ are finite,

$$(26) \quad \left(\int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \lesssim_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

For the converse estimate, let P_g denote the orthogonal projection of $\mathbb{C}^G$ onto the g -th coordinate and define

$$\mathcal{E}(A) := \sum_{g \in G} P_g A P_g, \quad A \in \mathcal{L}(\mathbb{C}^G).$$

Thus, $\mathcal{E}(A)$ is the diagonal part of A . The pinching inequality for unitarily invariant norms [?, p. 97, (IV.51)–(IV.52)] gives

$$(27) \quad \|\mathcal{E}(A)\|_{S^N} \leq \|A\|_{S^N}.$$

For every $\alpha \in R_+$, S_α has zero diagonal, since $g\sigma_\alpha \neq g$ for all $g \in G$. Hence, by the definition of $\mathfrak{c}_{f,j}$,

$$\mathcal{E}(\mathfrak{c}_{f,j}(x, \theta)) = \text{diag}_{g \in G} \left(\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right).$$

Therefore, by (27),

$$(28) \quad \|\mathfrak{c}_{f,j}(x, \theta)\|_{S^N}^N \geq \sum_{g \in G} \left| \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right|^N.$$

For each $g \in G$, set

$$y := gx, \quad \eta := g\theta.$$

Since

$$\nabla f_g(x) = g^{-1} \nabla f(y), \quad d_{g,j} = g^{-1} e_j,$$

and g is orthogonal,

$$\begin{aligned} \langle d_{g,j}, \nabla f_g(x) \rangle &= \partial_j f(y), \\ \langle d_{g,j}, \theta \rangle &= \eta_j, \\ \langle \theta, \nabla f_g(x) \rangle &= \langle \eta, \nabla f(y) \rangle. \end{aligned}$$

Using the chamber decomposition and the orthogonal invariance of surface measure, (28) therefore gives

$$\begin{aligned} (29) \quad & \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \|\zeta_{f,j}(x, \theta)\|_{\mathbb{S}^N}^N d\theta dx \\ & \geq \int_{\mathbb{R}^N} \int_{\mathbb{S}^{N-1}} |\partial_j f(y) - \eta_j \langle \eta, \nabla f(y) \rangle|^N d\eta dy. \end{aligned}$$

For $v \in \mathbb{C}^N$, set

$$\Phi_j(v) := \left(\int_{\mathbb{S}^{N-1}} |v_j - \eta_j \langle \eta, v \rangle|^N d\eta \right)^{1/N}.$$

Then

$$(30) \quad \Phi_j(v) \simeq_N \|v\|.$$

Indeed, Φ_j is a seminorm on $\mathbb{C}^N$. If $\Phi_j(v) = 0$, continuity gives

$$v_j - \eta_j \langle \eta, v \rangle = 0, \quad \eta \in \mathbb{S}^{N-1}.$$

Since $N > 1$, there exists $\eta \in \mathbb{S}^{N-1}$ with $\eta_j = 0$, and therefore $v_j = 0$. It follows that

$$\langle \eta, v \rangle = 0$$

whenever $\eta_j \neq 0$. Since such η form a dense subset of $\mathbb{S}^{N-1}$, continuity yields $\langle \eta, v \rangle = 0$ for every $\eta \in \mathbb{S}^{N-1}$, and hence $v = 0$. Thus Φ_j is a norm, and (30) follows from the equivalence of norms on the finite dimensional space $\mathbb{C}^N$.

Applying (30) with $v = \nabla f(y)$ in (29), we obtain

$$\|\nabla f\|_{L^N(\mathbb{R}^N, dx)} \lesssim_N \left(\int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\zeta_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Combining this estimate with (26) proves the assertion. $\square$

Lemma 6.6. Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in C_c^\infty(\mathbb{R}^N)$. Define

$$A_f := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

Then there exist $\chi_m \in C_c^\infty(\Gamma_+)$, $0 \leq \chi_m \leq 1$, such that $\chi_m(x) \rightarrow 1$ for every $x \in \Gamma_+$ and

$$(31) \quad \lim_{m \rightarrow \infty} \|A_f - M_{\chi_m} A_f M_{\chi_m}\|_{S^{N,\infty}} = 0.$$

Proof. Fix $s > 2$, choose $2 < s_0 < s$, and set

$$\gamma := \frac{1}{s_0} - \frac{1}{s} > 0.$$

Let $r_s > s$ be as in Lemma ?? and set $\beta := N/r_s$. By Lemma 8.2 (1), $f \in OSC^{N,\infty}(\mathbb{R}^N)$, so Lemma ?? applies with $p = N$ and $q = \infty$.

Set

$$B_f := U[R_{\text{Dunkl},j}, M_f]U^{-1}.$$

Since $V = M_\rho$ commutes with scalar multiplication operators on Γ ,

$$(32) \quad \|A_f - M_\chi A_f M_\chi\|_{S^{N,\infty}} = \|B_f - M_\chi B_f M_\chi\|_{S^{N,\infty}}$$

for every bounded scalar multiplier χ .

For fixed $\sigma, \tau \in G$, $u, v \in \mathbb{Z}^N$, and $\varepsilon \in \{0, 1, 2, 3\}$, define

$$T_{u,v,\varepsilon}^{\sigma,\tau} := \sum_{I \in \mathbb{D}_\Gamma} A_{\sigma,\tau,I,\varepsilon}(f) \left\langle \cdot, \overline{H_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot)} \right\rangle_{L^2(\Gamma, dw)} E_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot).$$

By the operator series identification in the proof of Theorem ??, the (σ, τ) matrix entry of B_f is

$$\sum_{\varepsilon=0}^3 \sum_{u,v \in \mathbb{Z}^N} c_{u,v} T_{u,v,\varepsilon}^{\sigma,\tau},$$

with convergence in $S^{N,\infty}$. Moreover, by Lemma ?? (??)–(??),

$$(33) \quad \text{supp } E_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot) \subset I, \quad \text{supp } H_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot) \subset I^*,$$

$$(34) \quad \|E_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot)\|_{L^s(dw)} + \|H_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot)\|_{L^s(dw)} \lesssim w(I)^{1/s-1/2}.$$

The definitions of $A_{\sigma,\tau,I,\varepsilon}(f)$ in the proof of Lemma ?? imply

$$(35) \quad |A_{\sigma,\tau,I,\varepsilon}(f)| \lesssim_f \begin{cases} \ell(I), & \ell(I) \leq 1, \\ \ell(I)^{-\beta}, & \ell(I) \geq 1. \end{cases}$$

Indeed, for $\varepsilon = 0, 1, 2$, the corresponding coefficients are bounded by an $L^{r_s}(dx)$ oscillation of a smooth compactly supported function on a set of diameter comparable with $\ell(I)$. Hence the first estimate follows from the Lipschitz bound, while for $\ell(I) \geq 1$,

$$\text{osc}_{r_s}(g; E) \lesssim |E|^{-1/r_s} \|g\|_{L^{r_s}} \lesssim_f \ell(I)^{-N/r_s}.$$

For $\varepsilon = 3$, writing $D := |\sigma z_I - \tau z_I|$, the definitions of $\theta_{\sigma,\tau,I}$ and $a_{\sigma,\tau,I}(f)$ imply, for $\ell(I) \leq 1$,

$$|A_{\sigma,\tau,I,3}(f)| \lesssim_f \frac{\ell(I)}{\ell(I) + D} (\ell(I) + D) \lesssim_f \ell(I),$$

whereas, for $\ell(I) \geq 1$,

$$|A_{\sigma,\tau,I,3}(f)| \leq |I|^{-1} \|H_{\sigma,\tau}(f)\|_{L^1} \lesssim_f \ell(I)^{-N} \lesssim \ell(I)^{-\beta}.$$

Let $\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$. For $1 \leq i \leq N_0$ and $0 < \delta \leq 1$, set

$$W_{i,\delta} := \{x \in \Gamma : 0 \leq \langle x, \alpha_i \rangle \leq 2\delta\}.$$

We have

$$(36) \quad \frac{w(W_{i,\delta} \cap I)}{w(I)} + \frac{w(W_{i,\delta} \cap I^*)}{w(I)} \lesssim \min\left(1, \frac{\delta}{\ell(I)}\right).$$

To prove this, write $I = \Phi^{-1}(Q)$ and $\ell := \ell(I)$. The assertion is immediate when $\delta > \ell/8$, since $w(I^*) \lesssim w(I)$. Suppose that $\delta \leq \ell/8$. If I or I^* meets $W_{i,\delta}$, then the i -th coordinate interval of Q is $[0, \ell)$. By Lemma 3.9 (5), the density of $\Phi_\# dw$ is

$$\phi(u) = |\det(\Phi)|^{-1} \prod_{v \in R_+} \left(\sum_{h=1}^{N_0} a_{v,h} u_h \right)^{2k(v)},$$

which is nondecreasing in every u_i . If g is nonnegative and nondecreasing on an interval of length comparable with ℓ , then

$$\int_0^{2\delta} g(t) dt \lesssim \frac{\delta}{\ell} \int g(t) dt.$$

Applying this estimate fiberwise to ϕ on Q , respectively on $(2Q) \cap \Omega$, and using $w(I^*) \lesssim w(I)$, we obtain (36).

Define

$$\omega_{I,\delta} := \min \left(1, \left(\frac{\delta}{\ell(I)} \right)^\gamma \right).$$

If $0 \leq a \leq 1$ and $\text{supp } a \subset W_{i,\delta}$, then by Hölder's inequality, (34), and (36),

$$\|aE_{u,v,I,\varepsilon}^{\sigma,\tau}(\sigma \cdot)\|_{L^{s_0}(dw)} + \|aH_{u,v,I,\varepsilon}^{\sigma,\tau}(\tau \cdot)\|_{L^{s_0}(dw)} \lesssim \omega_{I,\delta} w(I)^{1/s_0-1/2}.$$

After normalization by $\omega_{I,\delta}$, the localized factors are NWO families with exponent s_0 . Hence, by Lemma 2.10,

$$(37) \quad \|M_a T_{u,v,\varepsilon}^{\sigma,\tau}\|_{S^{N,\infty}} + \|T_{u,v,\varepsilon}^{\sigma,\tau} M_a\|_{S^{N,\infty}} \lesssim \left\| \left\{ A_{\sigma,\tau,I,\varepsilon}(f) \omega_{I,\delta} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N,\infty}}.$$

If $\ell(I) \leq 1$ and $A_{\sigma,\tau,I,\varepsilon}(f) \neq 0$, then z_I belongs to a fixed bounded set depending only on f . Consequently, at every dyadic length $0 < \ell \leq 1$,

$$(38) \quad \#\left\{ I \in \mathbb{D}_\Gamma : \ell(I) = \ell, A_{\sigma,\tau,I,\varepsilon}(f) \neq 0, (I \cup I^*) \cap W_{i,\delta} \neq \emptyset \right\} \lesssim_f \left(1 + \frac{\delta}{\ell} \right) \ell^{-(N-1)}.$$

Indeed, under Φ , there are $O(1 + \delta/\ell)$ admissible intervals in the u_i coordinate and $O(\ell^{-1})$ admissible intervals in each of the remaining $N-1$ coordinates.

For $\ell(I) \leq \delta$, (35) and (38) imply, for every $\lambda > 0$,

$$\begin{aligned} & \#\left\{ I : \ell(I) \leq \delta, |A_{\sigma,\tau,I,\varepsilon}(f) \omega_{I,\delta}| > \lambda \right\} \\ & \lesssim_f \delta \sum_{\substack{\ell \leq \delta \\ \lambda \lesssim_f \ell}} \ell^{-N} \lesssim_f \delta \lambda^{-N}. \end{aligned}$$

Therefore,

$$(39) \quad \left\| \left\{ A_{\sigma,\tau,I,\varepsilon}(f) \omega_{I,\delta} \right\}_{\ell(I) \leq \delta} \right\|_{\ell^{N,\infty}} \lesssim_f \delta^{1/N}.$$

For $\delta < \ell(I) \leq 1$,

$$|A_{\sigma,\tau,I,\varepsilon}(f) \omega_{I,\delta}| \lesssim_f \delta^\gamma \ell(I)^{1-\gamma},$$

and (38) reduces to $O_f(\ell^{-(N-1)})$. Given $\lambda > 0$, set

$$L := \max \left(\delta, c_f \left(\frac{\lambda}{\delta^\gamma} \right)^{1/(1-\gamma)} \right).$$

Since $N > 1$,

$$\#\left\{ I : \delta < \ell(I) \leq 1, |A_{\sigma,\tau,I,\varepsilon}(f) \omega_{I,\delta}| > \lambda \right\} \lesssim_f L^{-(N-1)}.$$

If $L = \delta$, then $\lambda \lesssim_f \delta$, and hence

$$\lambda L^{-(N-1)/N} \lesssim_f \delta^{1/N}.$$

Otherwise, $\lambda \lesssim_f \delta^\gamma L^{1-\gamma}$, so

$$\lambda L^{-(N-1)/N} \lesssim_f \delta^\gamma L^{1/N-\gamma} \lesssim_f \delta^{\min(\gamma, 1/N)}.$$

Thus,

$$(40) \quad \left\| \left\{ A_{\sigma, \tau, I, \varepsilon}(f) \omega_{I, \delta} \right\}_{\delta < \ell(I) \leq 1} \right\|_{\ell^{N, \infty}} \lesssim_f \delta^{\min(\gamma, 1/N)}.$$

For $\ell(I) \geq 1$, only $O_f(1)$ cubes at each dyadic scale have nonzero coefficients. By (35),

$$\sum_{\ell(I) \geq 1} |A_{\sigma, \tau, I, \varepsilon}(f) \omega_{I, \delta}|^N \lesssim_f \delta^{\gamma N} \sum_{\substack{\ell \geq 1 \\ \ell \text{ dyadic}}} \ell^{-N(\beta + \gamma)} \lesssim_f \delta^{\gamma N}.$$

Combining this with (39) and (40), we obtain

$$(41) \quad \left\| \left\{ A_{\sigma, \tau, I, \varepsilon}(f) \omega_{I, \delta} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{N, \infty}} \lesssim_f \delta^\kappa, \quad \kappa := \min\left(\gamma, \frac{1}{N}\right).$$

By (37),

$$(42) \quad \|M_a T_{u, v, \varepsilon}^{\sigma, \tau}\|_{S^{N, \infty}} + \|T_{u, v, \varepsilon}^{\sigma, \tau} M_a\|_{S^{N, \infty}} \lesssim_f \delta^\kappa$$

whenever $0 \leq a \leq 1$ and $\text{supp } a \subset W_{i, \delta}$.

We also need localization at infinity. Let $0 \leq b_R \leq 1$ with $\text{supp } b_R \subset \{x \in \Gamma : |x| \geq R\}$. If $A_{\sigma, \tau, I, \varepsilon}(f) \neq 0$, then

$$(43) \quad |z_I| \lesssim_f 1 + \ell(I).$$

For $\varepsilon = 0, 1$, the ball $B_\Gamma(z_I, C_1 \ell(I))$ must meet the support of $f \circ \tau$; for $\varepsilon = 2, 3$, I must meet the support of $H_{\sigma, \tau}(f)$. Since $|x - z_I| \lesssim \ell(I)$ for $x \in I^*$, (43) shows that, for sufficiently large R ,

$$A_{\sigma, \tau, I, \varepsilon}(f) \neq 0, \quad (I \cup I^*) \cap \{x : |x| \geq R\} \neq \emptyset$$

can occur only when $\ell(I) \gtrsim_f R$. At each such dyadic scale there are $O_f(1)$ cubes with nonzero coefficients. Hence, by (35) and Lemma 2.10,

$$(44) \quad \|M_{b_R} T_{u, v, \varepsilon}^{\sigma, \tau}\|_{S^{N, \infty}} + \|T_{u, v, \varepsilon}^{\sigma, \tau} M_{b_R}\|_{S^{N, \infty}} \lesssim_f R^{-\beta}.$$

Choose $\eta \in C_c^\infty(\mathbb{R})$, $0 \leq \eta \leq 1$, such that

$$\eta(t) = 0 \quad \text{for } t \leq 1, \quad \eta(t) = 1 \quad \text{for } t \geq 2,$$

and choose $\zeta \in C_c^\infty(\mathbb{R}^N)$, $0 \leq \zeta \leq 1$, such that

$$\zeta = 1 \quad \text{on } B(0, 1), \quad \text{supp } \zeta \subset B(0, 2).$$

For $0 < \delta \leq 1$ and $R \geq 1$, define

$$\chi_{\delta, R}(x) := \zeta(x/R) \prod_{i=1}^{N_0} \eta\left(\frac{\langle x, \alpha_i \rangle}{\delta}\right), \quad x \in \Gamma_+.$$

Then $\chi_{\delta, R} \in C_c^\infty(\Gamma_+)$ and $0 \leq \chi_{\delta, R} \leq 1$. Moreover,

$$\begin{aligned} 1 - \chi_{\delta, R} &= 1 - \zeta(\cdot/R) \\ &+ \sum_{i=1}^{N_0} \zeta(\cdot/R) \prod_{h < i} \eta\left(\frac{\langle \cdot, \alpha_h \rangle}{\delta}\right) \left(1 - \eta\left(\frac{\langle \cdot, \alpha_i \rangle}{\delta}\right)\right). \end{aligned}$$

The first term is supported in $\{x : |x| \geq R\}$, while the i -th term in the sum is supported in $W_{i,\delta}$. Since G is finite, $S^{N,\infty}$ is normable for $N > 1$, and Lemma ?? (??) implies

$$\sum_{u,v \in \mathbb{Z}^N} |c_{u,v}| < \infty,$$

we may sum (42) and (44) over the operator series for B_f . Therefore,

$$\|(I - M_{\chi_{\delta,R}})B_f\|_{S^{N,\infty}} + \|B_f(I - M_{\chi_{\delta,R}})\|_{S^{N,\infty}} \lesssim_f \delta^\kappa + R^{-\beta}.$$

Since

$$B_f - M_{\chi_{\delta,R}} B_f M_{\chi_{\delta,R}} = (I - M_{\chi_{\delta,R}})B_f + M_{\chi_{\delta,R}} B_f (I - M_{\chi_{\delta,R}}),$$

the ideal property of $S^{N,\infty}$ and (32) imply

$$\|A_f - M_{\chi_{\delta,R}} A_f M_{\chi_{\delta,R}}\|_{S^{N,\infty}} \lesssim_f \delta^\kappa + R^{-\beta}.$$

Define

$$\chi_m := \chi_{2^{-m}, 2^m}.$$

Then $\chi_m(x) \rightarrow 1$ for every $x \in \Gamma_+$, while

$$\|A_f - M_{\chi_m} A_f M_{\chi_m}\|_{S^{N,\infty}} \lesssim_f 2^{-m\kappa} + 2^{-m\beta} \rightarrow 0.$$

This completes the proof of Lemma 9.1. $\square$

Lemma 6.7. Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in C_c^\infty(\mathbb{R}^N)$. Then

$$(45) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Proof. Define

$$A_f := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

By Lemma 9.1 (2), there exist $\chi_m \in C_c^\infty(\Gamma_+)$, $0 \leq \chi_m \leq 1$, such that $\chi_m(x) \rightarrow 1$ for every $x \in \Gamma_+$ and

$$(46) \quad \|A_f - M_{\chi_m} A_f M_{\chi_m}\|_{S^{N,\infty}} \rightarrow 0.$$

Set

$$A_{f,m} := M_{\chi_m} A_f M_{\chi_m}, \quad K_m := \text{supp } \chi_m.$$

By Lemma 6.4 (17), $A_{f,m}$ is a compactly supported classical matrix valued pseudo-differential operator of order -1 on Γ_+ , with

$$(47) \quad \sigma_{-1}(A_{f,m})(x, \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathfrak{c}_{f,j}\left(x, \frac{\xi}{\|\xi\|}\right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

Choose a bounded open set U_m with $K_m \Subset U_m \Subset \Gamma_+$, and choose $L_m > 0$ such that

$$L_m > \sup_{x,y \in U_m} \|x - y\|_\infty.$$

For

$$\mathbb{T}_m^N := \mathbb{R}^N / L_m \mathbb{Z}^N,$$

the quotient map restricts to a diffeomorphism

$$\kappa_m : U_m \rightarrow \Omega_m := \kappa_m(U_m),$$

which is a local isometry. Since the Schwartz kernel of $A_{f,m}$ is supported in $K_m \times K_m$, with respect to

$$L^2(\Gamma; \mathbb{C}^G) = L^2(U_m; \mathbb{C}^G) \oplus L^2(\Gamma \setminus U_m; \mathbb{C}^G)$$

we have

$$A_{f,m} = B_{f,m} \oplus 0$$

for some $B_{f,m}$ on $L^2(U_m; \mathbb{C}^G)$.

Let

$$W_m : L^2(U_m; \mathbb{C}^G) \longrightarrow L^2(\Omega_m; \mathbb{C}^G), \quad W_m u := u \circ \kappa_m^{-1}.$$

Then W_m is unitary. By the invariance of classical pseudodifferential operators under smooth coordinate changes [?, Theorem 18.1.17],

$$T_{f,m} := W_m B_{f,m} W_m^{-1} \in \Psi_{\text{cl}}^{-1}(\Omega_m; \mathbb{C}^G).$$

Its Schwartz kernel is supported in $\kappa_m(K_m) \times \kappa_m(K_m) \Subset \Omega_m \times \Omega_m$. Choose $\vartheta_m \in C_c^\infty(\Omega_m)$ equal to 1 on a neighborhood of $\kappa_m(K_m)$. Then

$$T_{f,m} = M_{\vartheta_m} T_{f,m} M_{\vartheta_m}.$$

Hence the zero extension of the Schwartz kernel of $T_{f,m}$ from $\Omega_m \times \Omega_m$ to $\mathbb{T}_m^N \times \mathbb{T}_m^N$ is a classical pseudodifferential kernel of order -1 : locally in $\Omega_m \times \Omega_m$ this follows from $T_{f,m} \in \Psi_{\text{cl}}^{-1}$, while near the boundary of $\Omega_m \times \Omega_m$ the kernel vanishes. Denote the corresponding operator on $\mathbb{T}_m^N$ by $\widehat{A}_{f,m}$. Thus,

$$(48) \quad \widehat{A}_{f,m} \in \Psi_{\text{cl}}^{-1}(\mathbb{T}_m^N; \mathbb{C}^G).$$

Since κ_m is a local isometry, by (47) and the transformation law for principal symbols,

$$(49) \quad \sigma_{-1}(\widehat{A}_{f,m})(\kappa_m(x), \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathfrak{c}_{f,j} \left(x, \frac{\xi}{\|\xi\|} \right), \quad x \in U_m, \quad \xi \neq 0,$$

and the principal symbol vanishes outside Ω_m .

With respect to

$$L^2(\mathbb{T}_m^N; \mathbb{C}^G) = L^2(\Omega_m; \mathbb{C}^G) \oplus L^2(\mathbb{T}_m^N \setminus \Omega_m; \mathbb{C}^G),$$

we have

$$\widehat{A}_{f,m} = T_{f,m} \oplus 0.$$

Since $T_{f,m}$ and $B_{f,m}$ are unitarily equivalent,

$$(50) \quad \mu(t, \widehat{A}_{f,m}) = \mu(t, A_{f,m}), \quad t > 0.$$

By (48) and the Weyl law [?, Theorem 6.1],

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, \widehat{A}_{f,m}) = \left(\frac{1}{N(2\pi)^N} \int_{S^* \mathbb{T}_m^N} \text{tr}_{\mathbb{C}^G} |\sigma_{-1}(\widehat{A}_{f,m})(z, \theta)|^N dz d\theta \right)^{1/N}.$$

Since the metric on $\mathbb{T}_m^N$ is flat, combining (49) with (50), we obtain

$$(51) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_{f,m}) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N},$$

where χ_m is identified with its zero extension to Γ .

By Lemma 6.5,

$$(x, \theta) \longmapsto \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N$$

belongs to $L^1(\Gamma \times \mathbb{S}^{N-1})$. Since $0 \leq \chi_m \leq 1$, $\chi_m \rightarrow 1$ on Γ_+ , and $\Gamma \setminus \Gamma_+$ is contained in finitely many hyperplanes, the dominated convergence theorem implies

$$(52) \quad \begin{aligned} & \lim_{m \rightarrow \infty} \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \operatorname{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \\ &= \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \operatorname{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}. \end{aligned}$$

By (46), $A_{f,m} \rightarrow A_f$ in $S^{N,\infty}$. Combining (51), (52), and Lemma 6.8, we obtain

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_f) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \operatorname{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Since U and V are unitary,

$$\mu(t, A_f) = \mu(t, [R_{\text{Dunkl},j}, M_f]), \quad t > 0.$$

This completes the proof of Lemma 6.7. $\square$

Lemma 6.8. Let $0 < p < \infty$, let $\mathcal{H}$ be a Hilbert space, and let $A, A_m \in S^{p,\infty}(\mathcal{H})$, $m \geq 1$, satisfy

$$\|A_m - A\|_{S^{p,\infty}} \rightarrow 0.$$

Suppose that, for every $m \geq 1$, the limit

$$c_m := \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A_m)$$

exists. Then both limits

$$\lim_{m \rightarrow \infty} c_m \quad \text{and} \quad \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A)$$

exist and

$$\lim_{t \rightarrow \infty} t^{1/p} \mu(t, A) = \lim_{m \rightarrow \infty} c_m.$$

Proof. The assertion is the Birman–Solomyak approximation lemma; see [?, Section 11.6]. $\square$

Lemma 6.9. Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Then there exists $f_m \in C_c^\infty(\mathbb{R}^N)$ such that

$$\begin{aligned} & \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \rightarrow 0, \\ & \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} \rightarrow 0, \\ & \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \rightarrow 0. \end{aligned}$$

Proof. By the density of $C_c^\infty(\mathbb{R}^N)$ in $\dot{W}^{1,N}(\mathbb{R}^N)$, choose $f_m \in C_c^\infty(\mathbb{R}^N)$ such that

$$\|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \rightarrow 0.$$

By Theorem ?? and Lemma 8.2,

$$\begin{aligned} & \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} = \|[R_{\text{Dunkl},j}, M_{f_m-f}]\|_{S^{N,\infty}} \\ & \lesssim \|f_m - f\|_{OS C^{N,\infty}(\mathbb{R}^N)} \lesssim \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \rightarrow 0. \end{aligned}$$

By the linearity of $f \mapsto \mathfrak{c}_{f,j}$ and Lemma 6.5,

$$\begin{aligned} & \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} = \|\mathfrak{c}_{f_m-f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \\ & \lesssim_{N,R,k} \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \rightarrow 0. \end{aligned}$$

This completes the proof of Lemma 6.9. $\square$

Theorem 6.10. Assume that $N > 1$ and $j \in \{1, \dots, N\}$. Then, for every $f \in \dot{W}^{1,N}(\mathbb{R}^N)$,

$$(53) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\zeta_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Moreover,

$$(54) \quad \left(\int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\zeta_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

Proof. By Lemma 6.9, choose $f_m \in C_c^\infty(\mathbb{R}^N)$ such that

$$(55) \quad \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} \longrightarrow 0,$$

$$(56) \quad \|\zeta_{f_m,j} - \zeta_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow 0.$$

By Lemma 6.7, for every m ,

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_{f_m}]) = \frac{1}{(N(2\pi)^N)^{1/N}} \|\zeta_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

By (56),

$$\|\zeta_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow \|\zeta_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

Combining this convergence with (55), we apply Lemma 6.8 with $p = N$ and obtain (53). Finally, (54) follows from Lemma 6.5. This completes the proof of Theorem 6.10. $\square$

7. Alternative proof by using Dima's approach

Lemma 7.1. Fix a closed Weyl chamber Γ and $\sigma \in G \setminus \{\text{Id}\}$. Suppose that K satisfies

$$|\partial_x^\alpha \partial_y^\beta K(x, y)| \lesssim_{\alpha, \beta} \frac{\|x - y\|^{1-|\alpha|-|\beta|}}{\|x - \sigma y\| V(x, y, \|x - y\|)}$$

for every pair of multi-indices α, β and every $x, y \in \Gamma^\circ$ with $x \neq y$. For every $M_0 > 2N$, there exist positive numbers $\{c_{u,v}\}_{u,v \in \mathbb{Z}^N}$ and families $\{e_{u,v,I}\}_{u,v \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma}$ and $\{h_{u,v,I}\}_{u,v \in \mathbb{Z}^N, I \in \mathbb{D}_\Gamma}$ such that

$$(57) \quad |c_{u,v}| \lesssim_{M_0} (1 + |u| + |v|)^{-M_0},$$

$$(58) \quad \text{supp } e_{u,v,I} \subset I, \quad \text{supp } h_{u,v,I} \subset I^*,$$

$$(59) \quad |e_{u,v,I}| + |h_{u,v,I}| \lesssim w(I)^{-1/2},$$

and, for almost every $x, y \in \Gamma$ with $x \neq y$,

$$(60) \quad K(x, y) = \sum_{u,v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_\Gamma} \theta_{\text{Id}, \sigma, I} e_{u,v,I}(x) h_{u,v,I}(y).$$

Proof. Choose $\psi \in C_c^\infty(\mathbb{R}^N)$ such that

$$(61) \quad \text{supp } \psi \subset \left\{ z : \frac{1}{8} \leq |z| \leq \frac{1}{2} \right\}, \quad \sum_{m \in \mathbb{Z}} \psi(2^m z) = 1, \quad z \neq 0.$$

For $x, y \in \Gamma^\circ$, $x \neq y$, the unique cube $I \in \mathbb{D}_{\Gamma,m}$ containing x at each scale satisfies

$$(62) \quad K(x, y) = \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{\ell(I)}\right).$$

If the I -th summand is nonzero, then $|\Phi(x) - \Phi(y)| \leq \ell(I)/2$; hence $y \in I^*$.

Fix $I = \Phi^{-1}(Q)$. For

$$a \in (-1/2, 1/2)^N, \quad c_Q + \ell(I)b \in (2Q) \cap \Omega,$$

set

$$X := \Phi^{-1}(c_Q + \ell(I)a), \quad Y := \Phi^{-1}(c_Q + \ell(I)b).$$

Whenever $\psi(a-b) \neq 0$, by the G -invariance of d and dw , (4), and Lemma ?? applied with (Id, σ) ,

$$\begin{aligned} \|X - Y\| &\simeq \ell(I), \\ \|X - \sigma Y\| &\simeq \ell(I) + |z_I - \sigma z_I|, \\ (63) \quad V(X, Y, \|X - Y\|) &\simeq w(I). \end{aligned}$$

Consequently,

$$(64) \quad \left| \partial_x^\alpha \partial_y^\beta K(X, Y) \right| \lesssim_{\alpha, \beta} \theta_{\text{Id}, \sigma, I} \frac{\ell(I)^{-|\alpha| - |\beta|}}{w(I)}.$$

Let

$$\mathcal{R}_Q := \left\{ (a, b) : a \in (-1/2, 1/2)^N, \ c_Q + \ell(I)b \in (2Q) \cap \Omega \right\}.$$

On $\mathcal{R}_Q$, define

$$F_I(a, b) := \frac{w(I)}{\theta_{\text{Id}, \sigma, I}} \psi(a - b) K\left(\Phi^{-1}(c_Q + \ell(I)a), \Phi^{-1}(c_Q + \ell(I)b)\right).$$

By (64) and the chain rule, for every integer $L \geq 0$,

$$(65) \quad \sup_{I \in \mathbb{D}_\Gamma} \|F_I\|_{W^{L, \infty}(\mathcal{R}_Q)} \lesssim_L 1.$$

The rectangles $\mathcal{R}_Q$ range over finitely many fixed rectangular domains. Fix an integer $L > M_0$. By the standard extension theorem for Sobolev spaces on rectangles, followed by multiplication by a fixed smooth cutoff, F_I admits an extension $\widetilde{F}_I$ supported in $(-2, 2)^N \times (-2, 2)^N$ such that

$$(66) \quad \widetilde{F}_I = F_I \quad \text{on } \mathcal{R}_Q, \quad \sup_{I \in \mathbb{D}_\Gamma} \|\widetilde{F}_I\|_{W^{L, \infty}(\mathbb{R}^{2N})} \lesssim_L 1.$$

The 4-periodic extension of $\widetilde{F}_I$ has a Fourier expansion

$$\widetilde{F}_I(a, b) = \sum_{u, v \in \mathbb{Z}^N} b_{u, v, I} e^{\frac{\pi i}{2} u \cdot a} e^{\frac{\pi i}{2} v \cdot b}.$$

By repeated integration by parts and (66),

$$\sup_{I \in \mathbb{D}_\Gamma} |b_{u, v, I}| \lesssim_L (1 + |u| + |v|)^{-L}.$$

Increasing the implicit constant if necessary, define

$$c_{u, v} := C_{M_0} (1 + |u| + |v|)^{-M_0}$$

so that $|b_{u, v, I}| \leq c_{u, v}$ for every I . For $I = \Phi^{-1}(Q)$, define

$$\begin{aligned} e_{u, v, I}(x) &:= \frac{b_{u, v, I}}{c_{u, v}} w(I)^{-1/2} \chi_I(x) \exp\left(\frac{\pi i}{2\ell(I)} u \cdot (\Phi(x) - c_Q)\right), \\ h_{u, v, I}(y) &:= w(I)^{-1/2} \chi_{I^*}(y) \exp\left(\frac{\pi i}{2\ell(I)} v \cdot (\Phi(y) - c_Q)\right). \end{aligned}$$

Then (57)–(59) hold.

If $x \in I$ and $y \in I^*$, then

$$a := \frac{\Phi(x) - c_Q}{\ell(I)}, \quad b := \frac{\Phi(y) - c_Q}{\ell(I)}$$

belong to $\mathcal{R}_Q$. Hence, by (66) and the Fourier expansion,

$$(67) \quad K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{\ell(I)}\right) = \sum_{u, v \in \mathbb{Z}^N} c_{u, v} \theta_{\text{Id}, \sigma, I} e_{u, v, I}(x) h_{u, v, I}(y).$$

If $x \in I$ and $y \notin I^*$, both sides of (67) vanish. Therefore,

$$\chi_I(x) K(x, y) \psi\left(\frac{\Phi(x) - \Phi(y)}{\ell(I)}\right) = \sum_{u, v \in \mathbb{Z}^N} c_{u, v} \theta_{\text{Id}, \sigma, I} e_{u, v, I}(x) h_{u, v, I}(y)$$

for almost every $x, y \in \Gamma$. Summing over $I \in \mathbb{D}_\Gamma$ and using (61), we obtain (60).

This completes the proof of Lemma 7.1. $\square$

Lemma 7.2. Assume that $N \geq 1$, $1 < p < \infty$, and $1 \leq q \leq \infty$. Fix $s > 2$, a closed Weyl chamber Γ , and $\sigma \in G$. Let r_s be as in Lemma ???. Then, for every $f \in OSC^{p, q}(\mathbb{R}^N)$,

$$(68) \quad \left\| \left\{ \theta_{\sigma, \text{Id}, I} \text{osc}_{r_s}(H_{\sigma, \text{Id}}(f); I) \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p, q}} + \left\| \left\{ \theta_{\sigma, \text{Id}, I} a_{\sigma, \text{Id}, I}(f) \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p, q}} \lesssim \|f\|_{OSC^{p, q}(\mathbb{R}^N)}.$$

The implicit constant is independent of f and σ .

Proof. By (??), $0 < \theta_{\sigma, \text{Id}, I} \leq 1$. Hence, by Lemma 4.3,

$$\left\| \left\{ \theta_{\sigma, \text{Id}, I} \text{osc}_{r_s}(H_{\sigma, \text{Id}}(f); I) \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p, q}} \lesssim \left\| \left\{ \text{osc}_{r_s}(H_{\sigma, \text{Id}}(f); I) \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p, q}} \lesssim \|f\|_{OSC^{p, q}(\mathbb{R}^N)}.$$

By Lemma 4.2,

$$\left\| \left\{ \theta_{\sigma, \text{Id}, I} a_{\sigma, \text{Id}, I}(f) \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p, q}} \lesssim \|f\|_{OSC^{p, q}(\mathbb{R}^N)}.$$

Combining the two estimates, we obtain (68).

This completes the proof of Lemma 7.2. $\square$

Theorem 7.3. Assume that $N \geq 1$, $1 < p < \infty$, and $1 \leq q \leq \infty$. Fix a closed Weyl chamber Γ and $\sigma \in G \setminus \{\text{Id}\}$. Let T be an integral operator on $L^2(\Gamma, dw)$ whose kernel K satisfies

$$|\partial_x^\alpha \partial_y^\beta K(x, y)| \lesssim_{\alpha, \beta} \frac{\|x - y\|^{1-|\alpha|-|\beta|}}{\|x - \sigma y\| V(x, y, \|x - y\|)}$$

for every pair of multi-indices α, β and every $x, y \in \Gamma^\circ$ with $x \neq y$. Then, for every $f \in OSC^{p, q}(\mathbb{R}^N)$,

$$(69) \quad M_{f-f \circ \sigma} T \in S^{p, q}(L^2(\Gamma, dw)), \quad \|M_{f-f \circ \sigma} T\|_{S^{p, q}} \lesssim \|f\|_{OSC^{p, q}(\mathbb{R}^N)}.$$

Proof. Fix $s > 2$, let r_s be as in Lemma ??, and set

$$H := H_{\sigma, \text{Id}}(f) = f - f \circ \sigma.$$

For $I \in \mathbb{D}_\Gamma$, set

$$\omega_I := \text{osc}_{r_s}(H; I), \quad a_I := a_{\sigma, \text{Id}, I}(f) = (H)_{I, dx}.$$

By Lemma 7.2 (68),

$$(70) \quad \left\| \left\{ \theta_{\sigma, \text{Id}, I} \omega_I \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p, q}} + \left\| \left\{ \theta_{\sigma, \text{Id}, I} a_I \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{\ell^{p, q}} \lesssim \|f\|_{OSC^{p, q}(\mathbb{R}^N)}.$$

In particular, $\omega_I < \infty$ for every I . By Lemma ??, $H \in L^s_{\text{loc}}(\Gamma, dw)$.

Fix $M_0 > 2N$. By Lemma 7.1, for almost every $x, y \in \Gamma$ with $x \neq y$,

$$(71) \quad K(x, y) = \sum_{u, v \in \mathbb{Z}^N} c_{u, v} \sum_{I \in \mathbb{D}_\Gamma} \theta_{\text{Id}, \sigma, I} e_{u, v, I}(x) h_{u, v, I}(y),$$

where

$$\sum_{u, v \in \mathbb{Z}^N} |c_{u, v}| < \infty.$$

By (??), $\theta_{\text{Id}, \sigma, I} = \theta_{\sigma, \text{Id}, I}$.

For $u, v \in \mathbb{Z}^N$ and $I \in \mathbb{D}_\Gamma$, define

$$\tilde{e}_{u, v, I}(x) := \begin{cases} \omega_I^{-1} (H(x) - a_I) e_{u, v, I}(x), & \omega_I > 0, \\ 0, & \omega_I = 0. \end{cases}$$

By Lemma 7.1 (59) and Lemma ??,

$$\|\tilde{e}_{u, v, I}\|_{L^s(dw)} \lesssim w(I)^{-1/2} \omega_I^{-1} \left(\int_I |H - a_I|^s dw \right)^{1/s} \lesssim w(I)^{\frac{1}{s} - \frac{1}{2}}$$

when $\omega_I > 0$, while the same estimate is immediate when $\omega_I = 0$. Moreover, by Lemma 7.1 (58)–(59) and $w(I^*) \lesssim w(I)$,

$$\|e_{u, v, I}\|_{L^s(dw)} + \|h_{u, v, I}\|_{L^s(dw)} \lesssim w(I)^{\frac{1}{s} - \frac{1}{2}}.$$

Thus, for fixed u, v ,

$$\{\tilde{e}_{u, v, I}\}_{I \in \mathbb{D}_\Gamma}, \quad \{e_{u, v, I}\}_{I \in \mathbb{D}_\Gamma}, \quad \{\overline{h_{u, v, I}}\}_{I \in \mathbb{D}_\Gamma}$$

are NWO families.

If $\omega_I = 0$, then $H = a_I$ dx -a.e. on I , and hence dw -a.e. on I . Therefore, in all cases,

$$(72) \quad \begin{aligned} & \theta_{\sigma, \text{Id}, I} H(x) e_{u, v, I}(x) h_{u, v, I}(y) \\ &= \theta_{\sigma, \text{Id}, I} \omega_I \tilde{e}_{u, v, I}(x) h_{u, v, I}(y) + \theta_{\sigma, \text{Id}, I} a_I e_{u, v, I}(x) h_{u, v, I}(y). \end{aligned}$$

Hence, by Lemma 2.10 and (70),

$$(73) \quad \left\| \sum_{I \in \mathbb{D}_\Gamma} \theta_{\sigma, \text{Id}, I} \omega_I \langle \cdot, \overline{h_{u, v, I}} \rangle \tilde{e}_{u, v, I} + \sum_{I \in \mathbb{D}_\Gamma} \theta_{\sigma, \text{Id}, I} a_I \langle \cdot, \overline{h_{u, v, I}} \rangle e_{u, v, I} \right\|_{S^{p, q}} \lesssim \|f\|_{\text{OSC}^{p, q}(\mathbb{R}^N)},$$

uniformly in u, v .

Since $p > 1$, using an equivalent norm on $S^{p, q}$ when necessary, the series

$$(74) \quad S_f := \sum_{u, v \in \mathbb{Z}^N} c_{u, v} \sum_{I \in \mathbb{D}_\Gamma} \theta_{\sigma, \text{Id}, I} \left(\omega_I \langle \cdot, \overline{h_{u, v, I}} \rangle \tilde{e}_{u, v, I} + a_I \langle \cdot, \overline{h_{u, v, I}} \rangle e_{u, v, I} \right)$$

converges in $S^{p, q}(L^2(\Gamma, dw))$. By (73),

$$(75) \quad \|S_f\|_{S^{p, q}} \lesssim \sum_{u, v \in \mathbb{Z}^N} |c_{u, v}| \|f\|_{\text{OSC}^{p, q}(\mathbb{R}^N)} \lesssim \|f\|_{\text{OSC}^{p, q}(\mathbb{R}^N)}.$$

We identify S_f with $M_H T$. Fix $r > p$. By $\ell^{p, q} \subset \ell^r$ and (70),

$$\left\{ \theta_{\sigma, \text{Id}, I} \omega_I \right\}_{I \in \mathbb{D}_\Gamma}, \quad \left\{ \theta_{\sigma, \text{Id}, I} a_I \right\}_{I \in \mathbb{D}_\Gamma} \in \ell^r.$$

For $m \in \mathbb{N}$, let $S_f^{(m)}$ be defined by (74) with the sum over I restricted to $\bigcup_{|k| \leq m} \mathbb{D}_{\Gamma,k}$. By Lemma 2.10, now with exponent r ,

$$\begin{aligned} \|S_f - S_f^{(m)}\|_{S^r} &\lesssim \sum_{u,v \in \mathbb{Z}^N} |c_{u,v}| \left(\left\| \left\{ \theta_{\sigma, \text{Id}, I} \omega_I \right\}_{I \in \bigcup_{|k| > m} \mathbb{D}_{\Gamma,k}} \right\|_{\ell^r} \right. \\ &\quad \left. + \left\| \left\{ \theta_{\sigma, \text{Id}, I} a_I \right\}_{I \in \bigcup_{|k| > m} \mathbb{D}_{\Gamma,k}} \right\|_{\ell^r} \right) \rightarrow 0. \end{aligned}$$

Let $\varphi \in C_c^\infty(\Gamma^\circ)$ and

$$\eta \in C_c^\infty(\Gamma^\circ \setminus \{x : \sigma x = x\}).$$

By the construction in the proof of Lemma 7.1 and (72),

$$\begin{aligned} \langle S_f^{(m)} \varphi, \eta \rangle &= \int_\Gamma \int_\Gamma H(x) K(x, y) \left(\sum_{|k| \leq m} \psi(2^k(\Phi(x) - \Phi(y))) \right) \\ (76) \quad &\quad \times \varphi(y) \overline{\eta(x)} dw(y) dw(x). \end{aligned}$$

Set

$$\delta := \inf_{x \in \text{supp } \eta} \|x - \sigma x\| > 0.$$

For $x \in \text{supp } \eta$ and $\|x - y\| < \delta/2$,

$$\|x - \sigma y\| \geq \|x - \sigma x\| - \|x - y\| \geq \frac{\delta}{2}.$$

Moreover, since $x, y \in \Gamma$, by (4) and the G -invariance of d ,

$$\|x - \sigma y\| \geq d(x, \sigma y) = d(x, y) = \|x - y\|.$$

Consequently, the kernel estimate and the doubling property imply

$$\int_{\substack{y \in \Gamma \\ \|x-y\| < \delta/2}} |K(x, y)| dw(y) \lesssim \sum_{j=1}^{\infty} \frac{2^{-j} \delta}{\delta} \frac{w(B(x, 2^{-j} \delta))}{w(B(x, 2^{-j-1} \delta))} \lesssim \sum_{j=1}^{\infty} 2^{-j} \lesssim 1$$

uniformly in $x \in \text{supp } \eta$. On $\text{supp } \eta \times \text{supp } \varphi$ with $\|x - y\| \geq \delta/2$, the same annular argument involves only finitely many scales. Hence,

$$(77) \quad \sup_{x \in \text{supp } \eta} \int_{\text{supp } \varphi} |K(x, y)| dw(y) < \infty.$$

Since $H \in L_{\text{loc}}^s(\Gamma, dw)$ with $s > 2$, $H\eta \in L^2(\Gamma, dw) \cap L^1(\Gamma, dw)$. The partial sums in (76) are uniformly bounded and converge to 1 for $x \neq y$. Therefore, by (77) and dominated convergence,

$$\begin{aligned} \lim_{m \rightarrow \infty} \langle S_f^{(m)} \varphi, \eta \rangle &= \int_\Gamma \int_\Gamma H(x) K(x, y) \varphi(y) \overline{\eta(x)} dw(y) dw(x) \\ &= \langle T\varphi, \overline{H}\eta \rangle = \langle M_H T\varphi, \eta \rangle. \end{aligned}$$

On the other hand, the convergence in S^r implies

$$\lim_{m \rightarrow \infty} \langle S_f^{(m)} \varphi, \eta \rangle = \langle S_f \varphi, \eta \rangle.$$

Since $\sigma \neq \text{Id}$, the fixed point space $\{x : \sigma x = x\}$ is a proper linear subspace and therefore has dw -measure zero. Thus, $C_c^\infty(\Gamma^\circ \setminus \{x : \sigma x = x\})$ is dense in $L^2(\Gamma, dw)$, and S_f is the bounded extension of $M_H T = M_{f \circ \sigma} T$. Combining this identification with (75), we obtain (69).

This completes the proof of Theorem 7.3. $\square$

8. Appendix

Lemma 8.1. Let $1 < s < \infty$, and let $\mathbb{D}^0$ be the standard dyadic grid in $\mathbb{R}^N$. Suppose that $F \in L^s(\mathbb{R}^N)$. Then there exists a constant $C > 0$ such that

$$\left\| \left\{ |Q|^{-1/s'} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s,\infty}} \leq C \|F\|_{L^s(\mathbb{R}^N)}.$$

Proof. Let $\mathcal{E} \subset \mathbb{D}^0$ be finite and set

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Since the dyadic cubes containing any fixed x are totally ordered by inclusion, for each $x \in \bigcup_{Q \in \mathcal{E}} Q$ we may choose a cube $Q_x \in \mathcal{E}$ of minimal volume such that $x \in Q_x$. Every $Q \in \mathcal{E}$ containing x is then a dyadic ancestor of Q_x . Since the volume increases by the factor 2^N at each generation,

$$G_{\mathcal{E}}(x) = \sum_{\substack{Q \in \mathcal{E} \\ x \in Q}} |Q|^{-1/s'} \leq |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \lesssim_{N,s} \sup_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Consequently,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} \lesssim_{N,s} \int_{\mathbb{R}^N} \sup_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx \leq \int_{\mathbb{R}^N} \sum_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx = \#\mathcal{E}.$$

and hence,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

By Hölder's inequality,

$$\sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx = \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

Fix $t > 0$ and let $\mathcal{E}$ be any finite subcollection of the dyadic cubes satisfying

$$|Q|^{-1/s'} \int_Q |F(x)| dx > t.$$

Then

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'},$$

so

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since $\mathcal{E}$ is arbitrary,

$$\#\left\{ Q \in \mathbb{D}^0 : |Q|^{-1/s'} \int_Q |F(x)| dx > t \right\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

The asserted weak ℓ^s estimate follows. This completes the proof of Lemma 8.1. $\square$

Lemma 8.2. Assume that $N > 1$. Then the following statements hold.

- (1) For every $f \in L^1_{\text{loc}}(\mathbb{R}^N)$, we have
$$(1) \quad \|\nabla f\|_{L^N(\mathbb{R}^N)} \simeq \|f\|_{\text{OSC}^{N,\infty}(\mathbb{R}^N)}.$$
- (2) If $1 \leq q < \infty$ and $f \in \text{OSC}^{N,q}(\mathbb{R}^N)$, then f is constant almost everywhere.

- (3) If $1 \leq p < N$, $1 \leq q \leq \infty$, and $f \in OSC^{p,q}(\mathbb{R}^N)$, then f is constant almost everywhere.

Proof. This is a well-known result due to Rochberg–Semmes. $\square$

9. Answer to Dima's question

Lemma 9.1. Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in C_c^\infty(\mathbb{R}^N)$. Let Γ_1 and Γ_2 be Weyl chambers, and denote their interiors by Γ_1° and Γ_2° , respectively. Then there exist

$$\varphi_m \in C_c^\infty(\Gamma_1^\circ), \quad \psi_m \in C_c^\infty(\Gamma_2^\circ), \quad 0 \leq \varphi_m, \psi_m \leq 1,$$

such that

$$\varphi_m \longrightarrow \chi_{\Gamma_1^\circ}, \quad \psi_m \longrightarrow \chi_{\Gamma_2^\circ}$$

pointwise on $\mathbb{R}^N$, and

$$(2) \quad \lim_{m \rightarrow \infty} \|M_{\varphi_m}[R_{\text{Dunkl},j}, M_f]M_{\psi_m} - M_{\chi_{\Gamma_1}}[R_{\text{Dunkl},j}, M_f]M_{\chi_{\Gamma_2}}\|_{S^{N,\infty}} = 0.$$

Proof. Set

$$T_f := [R_{\text{Dunkl},j}, M_f].$$

We apply Lemma ?? with the fixed closed chamber $\Gamma = \Gamma_1$. Choose $\tau_0 \in G$ such that

$$\Gamma_2 = \tau_0 \Gamma_1.$$

By Lemma 8.2, $f \in OSC^{N,\infty}(\mathbb{R}^N)$, and hence the operator-series identification in the proof of Theorem ?? gives, with convergence in $S^{N,\infty}$,

$$T_f = \sum_{\varepsilon=0}^3 \sum_{\sigma, \tau \in G} \sum_{u,v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_{\Gamma_1}} A_{\sigma, \tau, I, \varepsilon}(f) \left\langle \cdot, \overline{H_{u,v,I,\varepsilon}^{\sigma, \tau}} \right\rangle E_{u,v,I,\varepsilon}^{\sigma, \tau}.$$

By Lemma ?? (??),

$$\text{supp } E_{u,v,I,\varepsilon}^{\sigma, \tau} \subset \sigma I, \quad \text{supp } H_{u,v,I,\varepsilon}^{\sigma, \tau} \subset \tau I^*.$$

Since distinct open Weyl chambers are disjoint and their boundaries have dw -measure zero,

$$(3) \quad M_{\chi_{\Gamma_1}} T_f M_{\chi_{\Gamma_2}} = \sum_{\varepsilon=0}^3 \sum_{u,v \in \mathbb{Z}^N} c_{u,v} T_{u,v,\varepsilon},$$

where

$$T_{u,v,\varepsilon} := \sum_{I \in \mathbb{D}_{\Gamma_1}} A_{I,\varepsilon} \left\langle \cdot, \overline{H_{I,\varepsilon}} \right\rangle E_{I,\varepsilon},$$

and, for brevity,

$$A_{I,\varepsilon} := A_{\text{Id}, \tau_0, I, \varepsilon}(f), \quad E_{I,\varepsilon} := E_{u,v,I,\varepsilon}^{\text{Id}, \tau_0}, \quad H_{I,\varepsilon} := H_{u,v,I,\varepsilon}^{\text{Id}, \tau_0}.$$

Thus

$$(4) \quad \text{supp } E_{I,\varepsilon} \subset I, \quad \text{supp } H_{I,\varepsilon} \subset \tau_0 I^*,$$

$$(5) \quad \|E_{I,\varepsilon}\|_{L^s(dw)} + \|H_{I,\varepsilon}\|_{L^s(dw)} \lesssim w(I)^{1/s-1/2}$$

for every fixed $s > 2$.

Fix $2 < s_0 < s$ and set

$$\gamma := \frac{1}{s_0} - \frac{1}{s} > 0.$$

Let $r_s > s$ be as in Lemma ??, and set

$$\beta := \frac{N}{r_s} > 0.$$

Exactly as in the coefficient estimate in the proof of the present lemma in its previous formulation, the definitions of $A_{\text{Id}, \tau_0, I, \varepsilon}(f)$ give

$$(6) \quad |A_{I, \varepsilon}| \lesssim_f \begin{cases} \ell(I), & \ell(I) \leq 1, \\ \ell(I)^{-\beta}, & \ell(I) \geq 1. \end{cases}$$

For $\varepsilon = 0, 1, 2$, this follows from the $L^{r_s}(dx)$ -oscillation defining the corresponding coefficient: the small-scale bound follows from the Lipschitz regularity of the relevant smooth function, whereas

$$\text{osc}_{r_s}(g; E) \lesssim |E|^{-1/r_s} \|g\|_{L^{r_s}} \lesssim_f \ell(I)^{-N/r_s}$$

when $\ell(I) \geq 1$. For $\varepsilon = 3$, if $D = |z_I - \tau_0 z_I|$, then for $\ell(I) \leq 1$,

$$|A_{I, 3}| \lesssim_f \frac{\ell(I)}{\ell(I) + D} (\ell(I) + D) \lesssim_f \ell(I),$$

while, for $\ell(I) \geq 1$,

$$|A_{I, 3}| \lesssim_f \ell(I)^{-N} \lesssim \ell(I)^{-\beta}.$$

Let

$$\Delta_1 = \{\alpha_1, \dots, \alpha_{N_0}\}$$

be the simple system associated with Γ_1 . For $1 \leq i \leq N_0$ and $0 < \delta \leq 1$, set

$$W_{i, \delta} := \{x \in \Gamma_1 : 0 \leq \langle x, \alpha_i \rangle \leq 2\delta\}.$$

As in the wall estimate in the previous proof,

$$(7) \quad \frac{w(W_{i, \delta} \cap I)}{w(I)} + \frac{w(W_{i, \delta} \cap I^*)}{w(I)} \lesssim \min\left(1, \frac{\delta}{\ell(I)}\right).$$

Indeed, the assertion is immediate for $\delta > \ell(I)/8$. If $\delta \leq \ell(I)/8$, write $I = \Phi^{-1}(Q)$. Whenever I or I^* meets $W_{i, \delta}$, the relevant i -th coordinate interval begins at the boundary of Ω . By Lemma 3.9 (5), the density of $\Phi_{\#}dw$ is nondecreasing in this coordinate. Hence the elementary estimate

$$\int_0^{2\delta} g(t) dt \lesssim \frac{\delta}{\ell(I)} \int g(t) dt$$

for nonnegative nondecreasing g , applied fiberwise on Q and $(2Q) \cap \Omega$, yields (7).

The simple system associated with Γ_2 is

$$\Delta_2 = \{\tau_0 \alpha_1, \dots, \tau_0 \alpha_{N_0}\}.$$

Set

$$\widetilde{W}_{i, \delta} := \tau_0 W_{i, \delta} = \{y \in \Gamma_2 : 0 \leq \langle y, \tau_0 \alpha_i \rangle \leq 2\delta\}.$$

Since dw is G -invariant, (7) also gives the corresponding estimate for $\widetilde{W}_{i, \delta} \cap \tau_0 I^*$.

Set

$$\omega_{I, \delta} := \min\left(1, \left(\frac{\delta}{\ell(I)}\right)^\gamma\right).$$

If $0 \leq a \leq 1$ and $\text{supp } a \subset W_{i, \delta}$, then (5), (7), and Hölder's inequality give

$$\|a E_{I, \varepsilon}\|_{L^{s_0}(dw)} \lesssim \omega_{I, \delta} w(I)^{1/s_0 - 1/2}.$$

Likewise, if $0 \leq b \leq 1$ and $\text{supp } b \subset \widetilde{W}_{i,\delta}$, then

$$\|bH_{I,\varepsilon}\|_{L^{s_0}(dw)} \lesssim \omega_{I,\delta} w(I)^{1/s_0-1/2}.$$

Here we use Lemma 2.10 in its standard two-family form: the two NWO families may be adapted to two dyadic systems related by a fixed measure-preserving Weyl-group element. The proof of Lemma 2.10 is unchanged, since the NWO estimates for the two families are used separately. Consequently,

$$(8) \quad \|M_a T_{u,v,\varepsilon}\|_{S^{N,\infty}} + \|T_{u,v,\varepsilon} M_b\|_{S^{N,\infty}} \lesssim \left\| \{A_{I,\varepsilon} \omega_{I,\delta}\}_{I \in \mathcal{I}_{i,\delta}} \right\|_{\ell^{N,\infty}},$$

where $\mathcal{I}_{i,\delta}$ denotes the cubes for which the corresponding localized factor is nonzero. In particular,

$$I \in \mathcal{I}_{i,\delta} \implies (I \cup I^*) \cap W_{i,\delta} \neq \emptyset.$$

If $\ell(I) \leq 1$ and $A_{I,\varepsilon} \neq 0$, then z_I belongs to a fixed bounded set depending only on f . Therefore, for every dyadic $0 < \ell \leq 1$,

$$(9) \quad \#\{I : \ell(I) = \ell, A_{I,\varepsilon} \neq 0, (I \cup I^*) \cap W_{i,\delta} \neq \emptyset\} \lesssim_f \left(1 + \frac{\delta}{\ell}\right) \ell^{-(N-1)}.$$

Indeed, in the Φ -coordinates there are $O(1 + \delta/\ell)$ admissible intervals in the i -th coordinate and $O(\ell^{-1})$ admissible intervals in each of the remaining $N-1$ coordinates.

For $\ell(I) \leq \delta$, we have $\omega_{I,\delta} = 1$. Hence, by (6) and (9),

$$\begin{aligned} & \#\{I \in \mathcal{I}_{i,\delta} : \ell(I) \leq \delta, |A_{I,\varepsilon} \omega_{I,\delta}| > \lambda\} \\ & \lesssim_f \delta \sum_{\substack{\ell \leq \delta \\ \lambda \lesssim_f \ell}} \ell^{-N} \lesssim_f \delta \lambda^{-N}, \end{aligned}$$

and therefore

$$(10) \quad \left\| \{A_{I,\varepsilon} \omega_{I,\delta}\}_{I \in \mathcal{I}_{i,\delta}} \right\|_{\ell^{N,\infty}} \lesssim_f \delta^{1/N}.$$

If $\delta < \ell(I) \leq 1$, then

$$|A_{I,\varepsilon} \omega_{I,\delta}| \lesssim_f \delta^\gamma \ell(I)^{1-\gamma},$$

whereas (9) reduces to $O_f(\ell^{-(N-1)})$. Given $\lambda > 0$, put

$$L := \max\left(\delta, c_f \left(\frac{\lambda}{\delta^\gamma}\right)^{1/(1-\gamma)}\right).$$

Then

$$\#\{I \in \mathcal{I}_{i,\delta} : \delta < \ell(I) \leq 1, |A_{I,\varepsilon} \omega_{I,\delta}| > \lambda\} \lesssim_f L^{-(N-1)}.$$

If $L = \delta$, then $\lambda \lesssim_f \delta$, and

$$\lambda L^{-(N-1)/N} \lesssim_f \delta^{1/N}.$$

Otherwise, $\lambda \lesssim_f \delta^\gamma L^{1-\gamma}$, whence

$$\lambda L^{-(N-1)/N} \lesssim_f \delta^\gamma L^{1/N-\gamma} \lesssim_f \delta^{\min(\gamma, 1/N)}.$$

Thus

$$(11) \quad \left\| \{A_{I,\varepsilon} \omega_{I,\delta}\}_{I \in \mathcal{I}_{i,\delta}} \right\|_{\ell^{N,\infty}} \lesssim_f \delta^{\min(\gamma, 1/N)}.$$

For $\ell(I) \geq 1$, only $O_f(1)$ cubes at each dyadic scale have nonzero coefficients. Hence

$$\sum_{\substack{I \in \mathcal{I}_{i,\delta} \\ \ell(I) \geq 1}} |A_{I,\varepsilon} \omega_{I,\delta}|^N \lesssim_f \delta^{\gamma N} \sum_{\substack{\ell \geq 1 \\ \text{dyadic}}} \ell^{-N(\beta+\gamma)} \lesssim_f \delta^{\gamma N}.$$

Combining this estimate with (10) and (11) gives

$$(12) \quad \left\| \{A_{I,\varepsilon} \omega_{I,\delta}\}_{I \in \mathcal{I}_{i,\delta}} \right\|_{\ell^{N,\infty}} \lesssim_f \delta^\kappa, \quad \kappa := \min\left(\gamma, \frac{1}{N}\right).$$

It follows from (8) that

$$(13) \quad \|M_a T_{u,v,\varepsilon}\|_{S^{N,\infty}} + \|T_{u,v,\varepsilon} M_b\|_{S^{N,\infty}} \lesssim_f \delta^\kappa$$

whenever

$$0 \leq a \leq 1, \quad \text{supp } a \subset W_{i,\delta},$$

or, respectively,

$$0 \leq b \leq 1, \quad \text{supp } b \subset \widetilde{W}_{i,\delta}.$$

We next treat localization at infinity. Suppose $0 \leq a_R \leq 1$ is supported in

$$\Gamma_1 \cap \{|x| \geq R\}.$$

If $A_{I,\varepsilon} \neq 0$, then, as in the previous proof,

$$|z_I| \lesssim_f 1 + \ell(I).$$

Since $E_{I,\varepsilon}$ is supported in I , for sufficiently large R ,

$$A_{I,\varepsilon} \neq 0, \quad I \cap \{|x| \geq R\} \neq \emptyset$$

can occur only if

$$\ell(I) \gtrsim_f R.$$

At every such scale there are $O_f(1)$ cubes with nonzero coefficients. Therefore

$$\|M_{a_R} T_{u,v,\varepsilon}\|_{S^{N,\infty}} \lesssim_f \left(\sum_{\ell(I) \gtrsim_f R} |A_{I,\varepsilon}|^N \right)^{1/N} \lesssim_f R^{-\beta}.$$

If $0 \leq b_R \leq 1$ is supported in

$$\Gamma_2 \cap \{|x| \geq R\},$$

then

$$\text{supp } H_{I,\varepsilon} \subset \tau_0 I^*$$

and $|\tau_0 x| = |x|$. The same argument gives

$$(14) \quad \|M_{a_R} T_{u,v,\varepsilon}\|_{S^{N,\infty}} + \|T_{u,v,\varepsilon} M_{b_R}\|_{S^{N,\infty}} \lesssim_f R^{-\beta}.$$

Choose $\eta \in C^\infty(\mathbb{R})$, $0 \leq \eta \leq 1$, such that

$$\eta(t) = 0 \quad \text{for } t \leq 1, \quad \eta(t) = 1 \quad \text{for } t \geq 2,$$

and choose a radial $\zeta \in C_c^\infty(\mathbb{R}^N)$, $0 \leq \zeta \leq 1$, satisfying

$$\zeta = 1 \quad \text{on } B(0, 1), \quad \text{supp } \zeta \subset B(0, 2).$$

For $0 < \delta \leq 1$ and $R \geq 1$, define on $\mathbb{R}^N$

$$\begin{aligned} \varphi_{\delta,R}(x) &:= \zeta(x/R) \prod_{i=1}^{N_0} \eta\left(\frac{\langle x, \alpha_i \rangle}{\delta}\right), \\ \psi_{\delta,R}(x) &:= \zeta(x/R) \prod_{i=1}^{N_0} \eta\left(\frac{\langle x, \tau_0 \alpha_i \rangle}{\delta}\right). \end{aligned}$$

Then

$$\varphi_{\delta,R} \in C_c^\infty(\Gamma_1^\circ), \quad \psi_{\delta,R} \in C_c^\infty(\Gamma_2^\circ), \quad 0 \leq \varphi_{\delta,R}, \psi_{\delta,R} \leq 1.$$

On Γ_1 ,

$$1 - \varphi_{\delta,R} = 1 - \zeta(\cdot/R) + \sum_{i=1}^{N_0} \zeta(\cdot/R) \prod_{h < i} \eta\left(\frac{\langle \cdot, \alpha_h \rangle}{\delta}\right) \left(1 - \eta\left(\frac{\langle \cdot, \alpha_i \rangle}{\delta}\right)\right).$$

The first term is supported in $\Gamma_1 \cap \{|x| \geq R\}$, and the i -th term in the sum is supported in $W_{i,\delta}$. The analogous decomposition for $1 - \psi_{\delta,R}$ on Γ_2 has its i -th wall term supported in $\widetilde{W}_{i,\delta}$.

Since $N > 1$, $S^{N,\infty}$ is normable. Moreover, Lemma ?? (??) gives

$$\sum_{u,v \in \mathbb{Z}^N} |c_{u,v}| < \infty.$$

Hence, summing (13) and (14) in (13), over ε, u, v , yields

$$\begin{aligned} & \|M_{\chi_{\Gamma_1} - \varphi_{\delta,R}} T_f M_{\chi_{\Gamma_2}}\|_{S^{N,\infty}} + \|M_{\chi_{\Gamma_1}} T_f M_{\chi_{\Gamma_2} - \psi_{\delta,R}}\|_{S^{N,\infty}} \\ (15) \quad & \lesssim_f \delta^\kappa + R^{-\beta}. \end{aligned}$$

Since

$$M_{\varphi_{\delta,R}} = M_{\varphi_{\delta,R}} M_{\chi_{\Gamma_1}}, \quad M_{\psi_{\delta,R}} = M_{\chi_{\Gamma_2}} M_{\psi_{\delta,R}},$$

we have

$$\begin{aligned} & M_{\chi_{\Gamma_1}} T_f M_{\chi_{\Gamma_2}} - M_{\varphi_{\delta,R}} T_f M_{\psi_{\delta,R}} \\ & = M_{\chi_{\Gamma_1} - \varphi_{\delta,R}} T_f M_{\chi_{\Gamma_2}} + M_{\varphi_{\delta,R}} T_f M_{\chi_{\Gamma_2} - \psi_{\delta,R}}. \end{aligned}$$

By the ideal property of $S^{N,\infty}$ and (15),

$$\|M_{\varphi_{\delta,R}} T_f M_{\psi_{\delta,R}} - M_{\chi_{\Gamma_1}} T_f M_{\chi_{\Gamma_2}}\|_{S^{N,\infty}} \lesssim_f \delta^\kappa + R^{-\beta}.$$

Finally, set

$$\varphi_m := \varphi_{2^{-m}, 2^m}, \quad \psi_m := \psi_{2^{-m}, 2^m}.$$

For every $x \in \Gamma_1^\circ$,

$$\varphi_m(x) \longrightarrow 1,$$

whereas $\varphi_m(x) = 0$ for every $x \notin \Gamma_1^\circ$. Thus

$$\varphi_m \longrightarrow \chi_{\Gamma_1^\circ}$$

pointwise. Similarly,

$$\psi_m \longrightarrow \chi_{\Gamma_2^\circ}$$

pointwise. Moreover,

$$\|M_{\varphi_m} T_f M_{\psi_m} - M_{\chi_{\Gamma_1}} T_f M_{\chi_{\Gamma_2}}\|_{S^{N,\infty}} \lesssim_f 2^{-m\kappa} + 2^{-m\beta} \longrightarrow 0.$$

This proves (2). $\square$

Lemma 9.2. Let Γ be a closed Weyl chamber, $\sigma \in G \setminus \{\text{Id}\}$, and let T satisfy the assumptions of Theorem 7.3. Assume $N > 1$ and $f \in C_c^\infty(\mathbb{R}^N)$. Set

$$H := f - f \circ \sigma.$$

Let

$$0 \leq f_m \leq 1, \quad f_m \in C_c^\infty(\Gamma^\circ),$$

and suppose that

$$f_m(x) \uparrow 1, \quad x \in \Gamma^\circ.$$

Then

$$\lim_{m \rightarrow \infty} \|M_{f_m H} T - M_H T\|_{S^{N,\infty}(L^2(\Gamma, dw))} = 0.$$

Proof. Fix $s > 2$, choose $2 < s_0 < s$, and set

$$\gamma := \frac{1}{s_0} - \frac{1}{s} > 0.$$

Let $r_s > s$ be as in Lemma ??, and put

$$\beta := \frac{N}{r_s}, \quad \kappa := \min\left(\gamma, \frac{1}{N}\right).$$

We first establish a localization estimate for $M_H T$. In the proof of Theorem 7.3, for fixed $u, v \in \mathbb{Z}^N$, one writes

$$M_H T = \sum_{\varepsilon=0}^1 \sum_{u,v \in \mathbb{Z}^N} c_{u,v} \sum_{I \in \mathbb{D}_\Gamma} A_{I,\varepsilon} \langle \cdot, \overline{F_{u,v,I,\varepsilon}} \rangle E_{u,v,I,\varepsilon},$$

with convergence in $S^{N,\infty}$, where

$$A_{I,0} = \theta_{\text{Id},\sigma,I} \text{osc}_{r_s}(H; I), \quad A_{I,1} = \theta_{\text{Id},\sigma,I}(H)_{I,dx},$$

and

$$\text{supp } E_{u,v,I,\varepsilon} \subset I, \quad \|E_{u,v,I,\varepsilon}\|_{L^s(dw)} + \|F_{u,v,I,\varepsilon}\|_{L^s(dw)} \lesssim w(I)^{1/s-1/2}.$$

We claim that

$$(16) \quad |A_{I,\varepsilon}| \lesssim_f \begin{cases} \ell(I), & \ell(I) \leq 1, \\ \ell(I)^{-\beta}, & \ell(I) \geq 1, \end{cases} \quad \varepsilon = 0, 1.$$

For $\varepsilon = 0$, this follows from the smoothness and compact support of H :

$$\text{osc}_{r_s}(H; I) \lesssim_f \ell(I), \quad \ell(I) \leq 1,$$

whereas

$$\text{osc}_{r_s}(H; I) \lesssim |I|^{-1/r_s} \|H\|_{L^s} \lesssim_f \ell(I)^{-\beta}, \quad \ell(I) \geq 1.$$

For $\varepsilon = 1$, put

$$D_I := |z_I - \sigma z_I|.$$

Since

$$H(z) = f(z) - f(\sigma z),$$

the Lipschitz regularity of f gives

$$|H(z)| \lesssim_f |z - \sigma z| \lesssim D_I + \ell(I), \quad z \in I.$$

Hence

$$|\theta_{\text{Id},\sigma,I}(H)_{I,dx}| \lesssim_f \frac{\ell(I)}{\ell(I) + D_I} (\ell(I) + D_I) \lesssim_f \ell(I).$$

For $\ell(I) \geq 1$,

$$|\theta_{\text{Id},\sigma,I}(H)_{I,dx}| \leq |I|^{-1} \|H\|_{L^1} \lesssim_f \ell(I)^{-N} \lesssim \ell(I)^{-\beta}.$$

Thus (16) holds.

Let

$$\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$$

be the simple system determining Γ , and set

$$W_{i,\delta} := \{x \in \Gamma : 0 \leq \langle x, \alpha_i \rangle \leq 2\delta\}.$$

The wall estimate used in the proof of Lemma 9.1 gives

$$(17) \quad \frac{w(W_{i,\delta} \cap I)}{w(I)} \lesssim \min\left(1, \frac{\delta}{\ell(I)}\right).$$

Consequently, if

$$0 \leq a \leq 1, \quad \text{supp } a \subset W_{i,\delta},$$

then Hölder's inequality gives

$$\|aE_{u,v,I,\varepsilon}\|_{L^{s_0}(dw)} \lesssim \omega_{I,\delta} w(I)^{1/s_0-1/2},$$

where

$$\omega_{I,\delta} := \min\left(1, \left(\frac{\delta}{\ell(I)}\right)^\gamma\right).$$

After normalization, Lemma 2.10 therefore yields

$$\left\| M_a \sum_{I \in \mathbb{D}_\Gamma} A_{I,\varepsilon} \langle \cdot, \overline{F_{u,v,I,\varepsilon}} \rangle E_{u,v,I,\varepsilon} \right\|_{S^{N,\infty}} \lesssim \left\| \{A_{I,\varepsilon} \omega_{I,\delta}\}_I \right\|_{\ell^{N,\infty}}.$$

Using (16) and exactly the dyadic counting argument in the proof of Lemma 9.1, one obtains

$$\left\| \{A_{I,\varepsilon} \omega_{I,\delta}\}_I \right\|_{\ell^{N,\infty}} \lesssim_f \delta^\kappa.$$

Since

$$\sum_{u,v \in \mathbb{Z}^N} |c_{u,v}| < \infty,$$

it follows that

$$(18) \quad \|M_a M_H T\|_{S^{N,\infty}} \lesssim_f \delta^\kappa$$

whenever a is supported in one of the strips $W_{i,\delta}$.

We also need localization at infinity. If

$$0 \leq a_R \leq 1, \quad \text{supp } a_R \subset \Gamma \cap \{|x| \geq R\},$$

then $A_{I,\varepsilon} \neq 0$ implies that I meets a fixed compact set depending only on f . Since $\text{supp } E_{u,v,I,\varepsilon} \subset I$, the additional condition

$$a_R E_{u,v,I,\varepsilon} \neq 0$$

forces

$$\ell(I) \gtrsim_f R.$$

There are only $O_f(1)$ such cubes at each dyadic scale, and (16) gives

$$(19) \quad \|M_{a_R} M_H T\|_{S^{N,\infty}} \lesssim_f R^{-\beta}.$$

Choose $\eta \in C^\infty(\mathbb{R})$, $0 \leq \eta \leq 1$, such that

$$\eta(t) = 0 \quad (t \leq 1), \quad \eta(t) = 1 \quad (t \geq 2),$$

and choose a radial $\zeta \in C_c^\infty(\mathbb{R}^N)$, $0 \leq \zeta \leq 1$, satisfying

$$\zeta = 1 \quad \text{on } B(0, 1), \quad \text{supp } \zeta \subset B(0, 2).$$

For $0 < \delta \leq 1$ and $R \geq 1$, set

$$\chi_{\delta,R}(x) := \zeta(x/R) \prod_{i=1}^{N_0} \eta\left(\frac{\langle x, \alpha_i \rangle}{\delta}\right), \quad x \in \Gamma.$$

Then

$$\chi_{\delta,R} \in C_c^\infty(\Gamma^\circ), \quad 0 \leq \chi_{\delta,R} \leq 1.$$

The function $1 - \chi_{\delta,R}$ is the sum of one function supported in

$$\Gamma \cap \{|x| \geq R\}$$

and finitely many functions supported in the strips $W_{i,\delta}$. Hence (18) and (19) imply

$$(20) \quad \|M_{1-\chi_{\delta,R}} M_H T\|_{S^{N,\infty}} \lesssim_f \delta^\kappa + R^{-\beta}.$$

We now return to the prescribed sequence f_m . Since

$$M_{f_m} H T - M_H T = -M_{1-f_m} M_H T,$$

it suffices to prove

$$\|M_{1-f_m} M_H T\|_{S^{N,\infty}} \longrightarrow 0.$$

Fix $\varepsilon > 0$. By (20), choose first $0 < \delta \leq 1$ and then $R \geq 1$ so that

$$\|M_{1-\chi_{\delta,R}} M_H T\|_{S^{N,\infty}} < \varepsilon.$$

Since

$$\text{supp } \chi_{\delta,R} \Subset \Gamma^\circ$$

and $f_m \uparrow 1$ pointwise on Γ° , Dini's theorem gives

$$\sup_{\text{supp } \chi_{\delta,R}} (1 - f_m) \longrightarrow 0.$$

Moreover,

$$1 - f_m = (1 - f_m)(1 - \chi_{\delta,R}) + (1 - f_m)\chi_{\delta,R}.$$

Therefore, by the ideal property of $S^{N,\infty}$,

$$\begin{aligned} \|M_{1-f_m} M_H T\|_{S^{N,\infty}} &\leq \|M_{1-f_m} M_{1-\chi_{\delta,R}} M_H T\|_{S^{N,\infty}} \\ &\quad + \|M_{(1-f_m)\chi_{\delta,R}} M_H T\|_{S^{N,\infty}} \\ &\leq \|M_{1-\chi_{\delta,R}} M_H T\|_{S^{N,\infty}} \\ &\quad + \sup_{\text{supp } \chi_{\delta,R}} (1 - f_m) \|M_H T\|_{S^{N,\infty}}. \end{aligned}$$

The first term is $< \varepsilon$, while the second tends to zero. Hence

$$\limsup_{m \rightarrow \infty} \|M_{1-f_m} M_H T\|_{S^{N,\infty}} \leq \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\|M_{1-f_m} M_H T\|_{S^{N,\infty}} \longrightarrow 0.$$

Equivalently,

$$M_{f_m(f-f \circ \sigma)} T \longrightarrow M_{f-f \circ \sigma} T$$

in $S^{N,\infty}(L^2(\Gamma, dw))$. □

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